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Paleomagnetism and geochronology of the Gwalior Sills, Bundelkhand craton, Northern India Block: New constraints on Greater India assembly

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ABSTRACT

We present an updated paleomagnetic pole from the Gwalior Sills in the Bundelkhand craton within the Northern India Block (NIB). Geochronological results from baddeleyite grains from one of the sills yielded an age of 1719 ± 7 Ma which together with a previously published age indicates the emplacement of sills between 1712 and 1756 Ma (~1730 Ma). The paleomagnetic pole calculated from additional sites in this study, combined with previous studies, falls at 13.5°N, 173.7°E (A95 = 3.6°, K = 98) indicating near equatorial latitudes for northern India. Limestone sampled a few meters above the contact with the sill exhibits similar directions consistent with having been baked by the sill. The pole does not resemble any younger poles from Peninsular India and receives a reliability score of R = 5. Dykes in the Singhbhum craton are slightly older (1765 Ma) and indicate low paleolatitudes for the Southern Indian Block (SIB). Although the Gwalior and Singhbhum poles data indicate low latitudes for both the NIB and SIB, they are statistically different and indicate that a rotation of at least 65° is required to bring the poles into accord. We propose that the NIB and SIB were in proximity but were separated by an ocean basin. We propose the name Gotosindhu ('Ancient Sea') for the body of water separating the NIB and SIB. We also review previous models for the assembly of the Columbia supercontinent during this time and critically examine the position of the NIB/SIB in those reconstructions.
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1. Introduction

Peninsular India comprises of at least six distinct ancient regions that include some of the oldest crust on the planet (Miller et al., 2018; Choudhuri et al., 2018; Bhattacharjee et al., 2021). These are the Bundelkhand and Aravalli cratons comprising the North India Block (NIB) and the Singhbhum, Bastar, and Dharwar cratons comprising the South India Block (SIB) (Fig. 1). The Marwar Craton is sometimes included in the NIB and other times treated as a separate entity, likely in part due to its unknown age (e.g., Meert et al., 2021; Ghosh et al., 2022). Portions of the Southern Granulite Province are attached to the southern edge of the Dharwar Craton and are considered part of the Dharwar Craton (Pivarunas et al., 2019). Paleomagnetic studies within India have advanced in the past twenty years (Meert et al., 2021), but several unknowns remain. When was the Peninsular India assembled into a single landmass? What is the history of that assembly? What role did India or its cratonic components play in the Precambrian supercontinents of Columbia and Rodinia?

The task of determining the geodynamic history of the blocks within Peninsular India is challenging due to the occurrence of multiple tectonic phases in each craton. The available geochronological data, although improving, remains limited in its coverage (Santosh et al., 2004; Meert and Pandit, 2015; Saha and Patranabis-Deb, 2014). The Dharwar and Bastar cratons are separated by the Prahnita-Godavari Rift (PG) and Bhopalpatnam Belt, and the Singhbhum and Bastar cratons are separated by the Mahanadi Rift (MR). The Central Indian Tectonic Zone (CITZ) is the fundamental structure that traverses Peninsular India from east to west and is thought to demarcate a major boundary between the NIB and the SIB. The CITZ is a polymetamorphic terrane with key metamorphic ages of ~2.5, ~1.6 and ~1.0 Ga (Bhowmik, 2019). Stein et al. (2004, 2006) maintained that the younger ~1.6 and ~1.0 Ga ages reflect reworking of material within an older ~2.5 Ga suture between the NIB and the SIB. Several authors find

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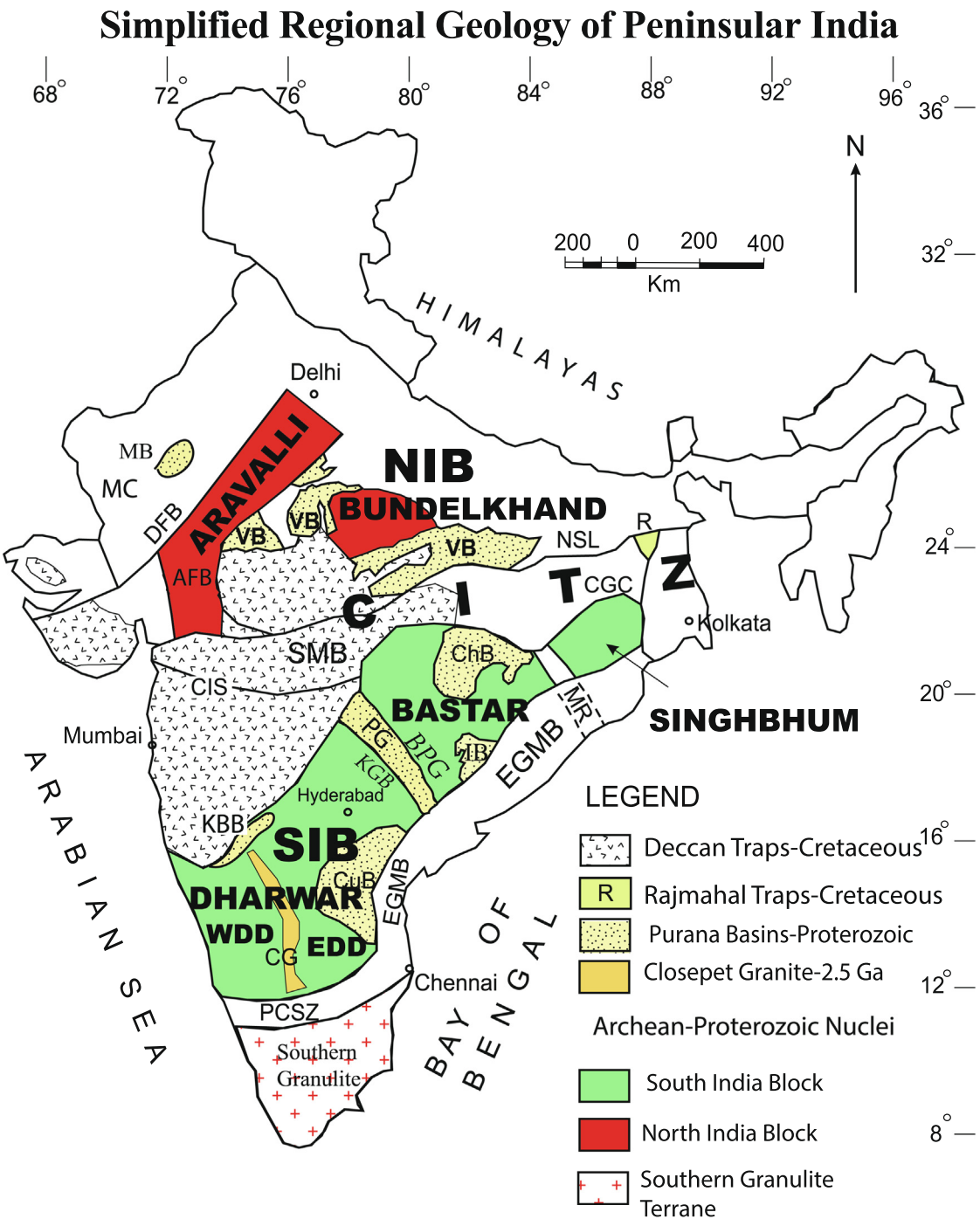


Fig. 1. Simplified regional map of Peninsular India (after Meert and Pandit, 2015). Abbreviations: MC = Mewar Craton; Orogenic Belts: CITZ = Central Indian Tectonic Zone; DFB = Delhi Fold Belt; AFB = Aravalli Fold Belt; NSL = Narmada-Son lineament; SMB = Satpura Mobile Belt; CIS = Central Indian Suture; CGC = Chottanagpur Gneissic Complex; BPG = Bhopalpatnam Granulite; KGB = Karimnagar Granulite Belt; EGMB = Eastern Ghats Mobile Belt; PCSZ = Palghat-Cauvery Shear Zone. Purana and other Basins; VB = Vindhyan Basin, MB = Marwar Basin; ChB = Chattisgarh Basin; IB = Indravati Basin; PG = Prahnita-Godavari Basin; CuB = Cuddapah Basin; KBB = Kaladgi-Bhima Basin; MR = Mahanadi Rift, NIB = North India Block; SIB = South India Block; WDD = Western Dharwar Domain; EDD = Eastern Dharwar Domain, CG = Closepet Granite; R = Rajmahal Traps.

the Stein et al. (2004, 2006) interpretation problematic and argue for a: (a) ~1.6 Ga collision between the SIB and the NIB (Bhandari et al., 2011); (b) a pre-Sausar collisional event unrelated to NIB/SIB amalgamation (Mandal et al., 2013); or (c) collision of several arc terranes separating with the SIB (Bhowmik, 2019). Others favor a much younger Stenian-Tonian (~1.1–0.88 Ga) assembly of the NIB, the SIB, and the Mewar (or Marwar) Craton to form 'Greater India' (Fig. 1; Bhowmik et al., 2012a, 2012b; Meert et al., 2013; Chatterjee, 2018; Meert et al., 2021; Ghosh et al., 2022). In this latter interpretation, the ~1.6 Ga orogenesis resulted from island arc collisions between the NIB and SIB prior to their final coalescence (Chatterjee, 2018; Bhowmik, 2019). Currently, existing proposals for both Rodinia and Columbia exhibit similarities in their core cratonic structures (Ernst et al., 2016; Meert, 2014; Evans and Mitchell, 2011; Meert and Santosh, 2017; Mitchell et al., 2021). However, these models might have

uncertainties due to the limited availability of reliable geochronological and paleomagnetic data. Uncertainties surround the sizes, compositions, and orientations of the constituent continents and cratons within these supercontinents. For instance, many iterations of these supercontinents depict Peninsular India as a unified entity rather than recognizing the separate elements of the NIB and the SIB. Additionally, India is often positioned along the margins of these supercontinents in a conventional East Gondwana configuration (adjacent to East Antarctica and Australia).

To better constrain Precambrian paleogeographic uncertainties for India, it is important to obtain additional high-quality paleomagnetic and geochronologic data. Meert et al. (2021) noted that a total of eleven reliable (with Q/R scores $\geq$5) paleomagnetic poles are available from Peninsular India which range in age from 2.37 to 0.77 Ga. Of these eleven poles, only two poles can ascertain India's position in Columbia and only one, controversial, grand mean pole is useful for determining India's position in Rodinia (Meert et al., 2021). In this paper, we report new paleomagnetic and geochronologic data from the Gwalior sills in the Bundelkhand craton and discuss their implications for paleogeographic reconstructions.

1.1. Bundelkhand craton geological history

The Bundelkhand craton is located to the east of the Aravalli Craton, both separated by the Great Boundary Fault (Fig. 1; also see Fig. 2b of Ghosh et al., 2022; Bickford et al., 2017; Mandal et al., 2018; Wiedenbeck et al., 1996). The Bundelkhand Craton contains typical cratonic rock assemblages such as Archean granitoids (and their gneissic equivalents) and greenstone belts intruded by Paleo-Mesoproterozoic mafic dyke swarms (Fig. 2a,b; e.g., Joshi et al., 2017; Slabunov and Singh, 2019; Colleps et al., 2021). Unique among Indian cratons, the Bundelkhand also hosts extensive NE-SW trending Paleoproterozoic quartz reefs (Pati et al., 2007). Among the mafic dykes in the Bundelkhand craton, Pradhan et al. (2012) provided U-Pb ages for an NW-SE trending swarm at 1979 ± 8 Ma and for younger, E-W dykes at 1113 ± 7 Ma. Pradhan et al. (2012) also noted a third suite of dykes based on distinct paleomagnetic directions but were unable to provide age constraints for them (see also Radhakrishna et al., 2013). Shankar et al. (2023) identified a 2004 ± 4 Ma suite of NW-SE trending dykes in the distal southeastern part of the Bundelkhand craton.

The narrow, E-W to ENE-WSE trending Paleoproterozoic Bijawar, Sonrai, and Gwalior basins overlie the basement rocks of the Bundelkhand craton (Absar et al., 2009; Chakraborty et al., 2015). The Bijawar and Sonrai basins contain similar lithologic successions known collectively as the Bijawar Group and are located along the southern margin of the craton (Fig. 2a; Dar et al., 2020). The Bijawar Group consists of two subgroups: the lower Moli Subgroup (conglomerate, mafic flows and sills, sandstones, and dolomites) and the upper Gangau Subgroup containing phosphorites and sandstone. The Gwalior basin is located in the northwestern part of the Bundelkhand craton and is ~25 km wide and 80 km long (Fig. 2a, b). The Gwalior Basin sedimentary rocks consist of two formations known together as the Gwalior Group (Absar et al., 2009). The basal Par Formation represents a continental to marine transgression in its lower ~60 m (Chakraborty et al., 2015). Much of the Par Formation is composed of shallow-water facies sandstones and shales. The overlying Morar Formation comprises shales, banded iron formations, ferruginous shales, and carbonates. The Gwalior Sills intrude interbedded shale, felsic tuff, and limestone at the contact between the Par and Morar formations and other locations within their stratigraphy (Chakraborty et al., 2015). For example, the Gwalior Sills also intrude iron formations at the top of the Morar Formation, lying stratigraphically beneath an unconformity and the Kaimur Group of the Vindhyan Supergroup (Varadarajan and Verma, 1971; Samom et al., 2018). The outcrop thickness of the sills ranges from a few meters (along road cuts) to more than 100 m in quarries. Borehole data reveal that the sills may be up to 500 m thick (Samom et al., 2018).

The ages of opening (initial deposition) and closure (final deposition) of the Gwalior and Bijawar basins, and of the mafic rocks that intrude them, are of continued interest. Absar et al. (2010) published Pb-Pb ages of 1866 ± 250 Ma and 1941 ± 120 Ma for carbonate units intruded by the Gwalior sills. Colleps et al. (2021) compiled detrital zircon ages from the Gwalior, Bijawar and Vindhyan basins to define the depositional and tectonic development of those basins and argued that sedimentation was initiated in both the Bijawar and Gwalior basins around 2.2 Ga. Their results indicate that the maximum age for the basal Gwalior and Bijawar sedimentation is ~2.15 Ga and the maximum age for the upper sequence of both basins is ~1.875 Ga. A Paleoproterozoic age for the Bijawar Basin closure is consistent with observations that show an unconformable onlap of the ~1.6–1.7 Ga Lower Vindhyan sediments onto the Bijawar sediments (Banerjee et al., 1982).

Rb-Sr and Pb-Pb ages for mafic intrusions into the Gwalior and Bijawar basins range from 1.79 to 1.69 Ga, although the ages often have greater than 0.1 Ga errors (Halder and Ghosh, 2000; Crawford and Compston, 1970). Samom et al. (2018) calculated a Sm-Nd whole-rock/mineral isochron age of 2104 ± 23 Ma for the Gwalior Sills. Although the error appears small, both the whole rock and plagioclase $^{147}Sm/^{144}Nd$ and $^{143}Nd/^{144}Nd$ ratios are nearly identical such that this is essentially a two-point isochron. A more recent in-situ dating of baddeleyite yielded a younger $^{207}Pb/^{206}Pb$ age of 1733 ± 23 Ma for the mafic sills in the Gwalior region (Srivastava et al., 2022). The 1.7–1.80 Ga magmatism is recognized in the Aravalli Craton (Biju-Sekhar et al., 2003, Kaur et al., 2011, Pandit et al., 2021) and large populations of 1.7–1.80 Ga detrital zircons are also found within the Upper Vindhyan and Marwar sedimentary rocks (Malone et al., 2008; McKenzie et al., 2011; Turner et al., 2014, Colleps et al., 2021; Zhang et al., 2022).

The tectonic setting for the opening of the Bijawar and Gwalior basins is uncertain. Colleps et al. (2021) argue that the basins formed via enhanced thermal subsidence following Bundelkhand craton stabilization. Chakraborty et al. (2015) argue that these represent rift basins associated with the breakup of the Kenorland super-cratons; however, the NIB was not included in the original proposals for Kenorland. Slabunov and Singh (2022), who suggested that the Bundelkhand craton was part of the Kenorland, argued that the basins formed during extension driven by a mantle plume; however, the only well-dated magmatic activity preceding basinal deposition in the Bundelkhand craton is ~1.98–2.0 Ga which is 150 million years younger than the proposed onset of Gwalior basin subsidence (Pradhan et al., 2012; Colleps et al., 2021; Shankar et al., 2023).

The tectonic setting of India during the emplacement of the Gwalior sills at ~1.73 Ga is equally controversial and overlaps with the opening of the Vindhyan Basin. The opening of the Vindhyan basin must be younger than 1.78–1.753 Ga based on the oldest zircon in the Semri Group (basal Vindhyan) and the age of the Jhiragandi granite onto which the lower Vindhyan sediments were deposited (Bora et al., 2013; Colleps et al., 2021). Closure of the Lower Vindhyan sedimentation is sometime after ~1.55 Ga (Meert and Pandit, 2015). The geochemical signatures of the Gwalior Sills suggest a within-plate extensional setting wherein the magma originated from an altered mantle source (Samom et al., 2018; Srivastava et al., 2022).

Bickford et al. (2017) provide a summary of the various tectonic models for Vindhyan basin development and conclude that some of the subsidence resulted from far-field extension related to collisional tectonics associated with Columbia assembly at ~1.65 Ga. They based their conclusions on the ages/geochemistry of the ~1.64 Ga rhyolite flows and lack of coeval mafic magmatism

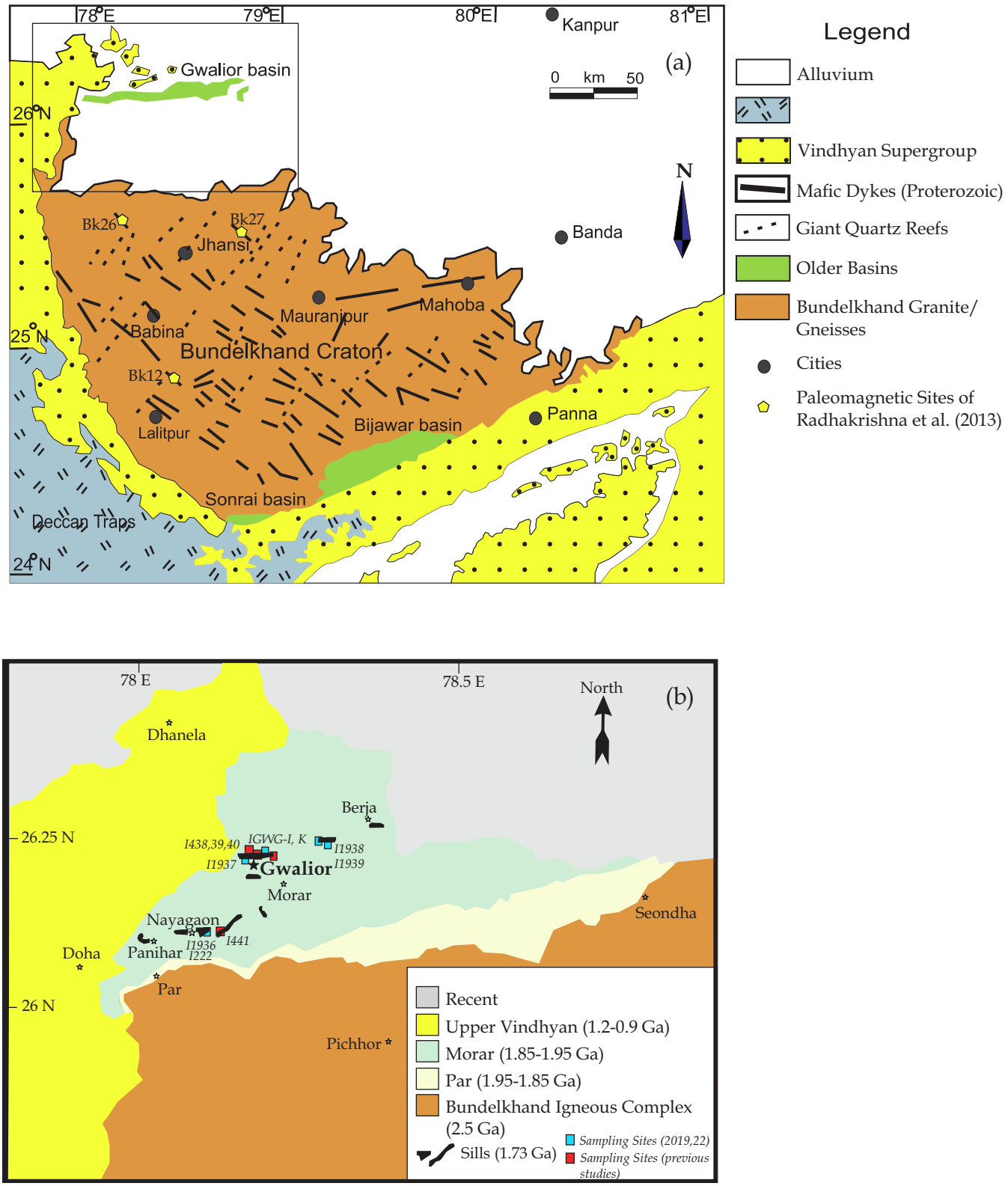


Fig. 2. (a) Generalized geological map of the Bundelkhand craton (after Malviya et al., 2004) showing the location of the Gwalior and Bijawar basins; (b) Gwalior basin showing sampling locations of the Gwalior sills in this study and previous studies (after Absar et al., 2009).

although a recent discovery by Mageswarii et al. (2021) has identified coeval mafic rocks within the Semri Group. The model of Bickford et al. (2017) does receive some support from the study of Bhowmik (2019) who argued that initial phases of collision in the region of the CITZ began sometime around 1.65–1.62 Ga with oceanic subduction beneath the SIB and ceased around 1.52 Ga after the collision of arc and back-arc systems beneath the SIB. The onset of subduction/collision overlaps with the 1.65–

1.593 Ga U-Pb ages determined on porcellanite (volcanic ash) sequences in the Semri Group; however, this does not explain the entire basin history which began nearly 100 million years earlier (Rasmussen et al., 2002; Ray et al., 2002,2003; Bengston et al., 2009; Bickford et al., 2017).

1.2. Previous paleomagnetic studies in the Bundelkhand craton

There are three Paleoproterozoic poles from dykes intruding the Bundelkhand craton (Poitou, 2009; Pradhan et al., 2010; Pradhan et al, 2012; Radhakrishna et al., 2013). Only one of the poles has a reliable U-Pb age of 1979 ± 8 Ma (Meert et al., 2021). A combined mean pole from the 1979 Ma dykes falls at 57.5°N, 309°E (A95 = 4.8°; Poitou, 2009; Pradhan et al., 2012; Radhakrishna et al., 2013). The dual-polarity magnetization has a baked contact test (Pradhan et al., 2012) demonstrating its primary nature and has an R = 6 according to Meert et al. (2020) criteria. Three other directions were isolated from dykes in the Bundelkhand craton (Pradhan et al., 2012; Radhakrishna et al., 2013). The steeply inclined directions are well-grouped, but the ages of the dykes yielding those data are unknown. Radhakrishna et al. (2013) assigned ages of 2.37 and 2.45 Ga to two of the steep directions (Steep 1 and Steep 2) based on comparisons with paleomagnetic directions observed in dykes from the Dharwar craton. These correlations are made without geochronological support. The Steep 1 direction (said to be 2.37 Ga) was isolated in 3 dykes in the Bundelkhand craton and yielded a paleomagnetic pole located at 10° S, 75°E (A95 = 18.1°) versus 12.8°N, 62°E (A95 = 4.6°) for the Dharwar mean (Belica et al., 2014; Meert et al., 2021). This direction conflicts with most of the well-dated 2.37 Ga dykes from the Dharwar craton that show reverse polarity (upwardly directed; see summary in Meert et al., 2021) because the three dykes from the Bundelkhand craton are of a single polarity (downwardly directed). Similar problems exist with the '2.45 Ga' dykes reported by Radhakrishna et al. (2013) and correlations between the Dharwar and Bundelkhand cratons are poorly known in the Paleo- and Mesoproterozoic. The ~2.0 Ga dykes dated by Shankar et al. (2023) have no associated paleomagnetic data; however, several dykes sampled by Radhakrishna et al. (2013) are some 15–35 km from the dated dykes. Two of the dyke directions correspond with those reported by Pradhan et al. (2012) for the 1.98 dykes and two dykes show distinct SW-shallow directions.

2. Paleomagnetic studies

2.1. Paleomagnetic methods

We sampled additional four sites from two sills near the city of Gwalior to complement previous studies. Site locations for this and previous sampling locations are shown in Fig. 2b. I1936 was collected at the upper level of the sill near the contact with sediments of the Morar Formation. Sites I1937, I1938 and I1939 were collected in the more massive parts of the sills. We also collected one site from the Morar limestones (I222) situated a few meters above the sill at Site I1936. Paleomagnetic samples were hand-drilled in the field using a portable gasoline-powered drill and oriented by magnetic compass and sun compass using a Pomeroy orientation stage (when the sun was available). Samples were returned to the University of Florida's Neil Opdyke Paleomagnetic Laboratory where they were cut into individual standard 2.5-cm size core specimens. When possible, samples were cut into two or more specimens. These duplicate specimens were used in pilot experiments to ascertain the best demagnetization method to isolate magnetic components. Samples were measured for bulk magnetic susceptibility on an AGICO KLY-3S instrument and then subjected to either stepwise thermal (ASC-TD-48) or alternating field demagnetization (Dtech-2000 demagnetizer) to isolate individual vector components using principal component analysis methods (Kirschvink, 1980) with IAPD software (Torsvik, 1986). All magnetic measurements were made on a Molspin Spinner magnetometer or 2G-77R cryogenic magnetometer. Temperature-susceptibility behavior was also tested on the AGICO KLY-3S (with CS-3 furnace attachment) and first-order reversal curves (FORCs) were run on a Princeton Instruments Micromag vibrating sample magnetometer (VSM) to help identify magnetic carriers.

2.2. Paleomagnetic results

Magnetic susceptibilities at site I1936 ranged from 0.5 to 8.0×10^{-3} (SI units). The other three sites yielded higher susceptibility values ranging from 12.1 to 60.0 $\times$ 10^{-3} (SI units). NRM intensities were variable amongst samples from I1936 (0.0165–1.600 A/m), but more consistent at Sites I1937 (0.76–1.1 A/m), I1938 (1.2–2.6 A/m) and I1939 (0.84–2.5 A/m).

Specimens from at least one sample from each site were demagnetized using both thermal and alternating field methods. Similar vectors were isolated using both thermal and alternating field demagnetization (Fig. 3a,b and Fig. 4a,b); however, thermal demagnetization was more effective in reducing the total NRM and was used in most samples. Typical demagnetization plots are shown in Figs. 3–4. Overprint directions were isolated at temperatures less than ~400–500 °C (Table 1). The mean angular deviation (MAD) for the characteristic directions isolated from individual samples ranged from 0.7° to 4.7° (Mean 2.3°).

Temperature-susceptibility was tested for representative specimens from each of the four sites (Fig. 5a–d). Site I1936 shows irreversible behavior indicating alteration of the magnetic mineralogy during heating. The estimated Curie temperatures are T_cH = 585 °C (heating) and T_cC = 588 °C (cooling). Site I1938 also shows irreversible behavior with two distinct Curie temperatures in the heating curve. The first T_cH = 340 °C and the second T_cH at 585 °C. The cooling curve shows a single T_cC = 580 °C. Samples from sites I1937 and I1939 show nearly reversible heating and cooling curves with Curie temps around 582 °C. Sample I1936-6 shows a large difference in susceptibility between the heating and cooling curves which is dominated by lower susceptibility during heating. The sample shows a sharp rise around the Curie temperature of magnetite (Hopkinson, 1889). Sample I1938-3 shows a drop in susceptibility around ~340 °C which may reflect the presence of pyrrhotite.

First-order reversal curves (FORCs) and isothermal remanent magnetization (IRM) curves were generated for selected samples from each of the four sites. Most samples show saturation by 0.3–0.5 T although a few show minor increases up to 0.5–1.0 T indicating a small contribution from the higher coercivity phases. Representative FORC diagrams are shown in Fig. 6a-d. There is some variation between the different sites, but all are consistent with interacting pseudo-single domain magnetite (Fig. 6a,b) with minor contributions from multi-domain grains (Fig. 6c,d; Egli, 2021).

Based on demagnetization (both AF and thermal), susceptibility-temperature behavior and rock magnetic characteristics, we conclude that SD-PSD titanomagnetite/magnetite grains dominate in most samples. This supports our interpretation of the high-temperature ChRM directions being of primary TRM origin. These grains can maintain their magnetization for billions of years and we see no rock-magnetic/outcrop evidence of subsequent alteration or reheating. Pyrrhotite is also present in some of the samples and is discussed below.

Table 1 lists paleomagnetic results from each of the 4 sites from this study along with previously published results (Athavale et al., 1963; Klootwijk, 1974; McElhinny et al., 1978, Pradhan et al.,

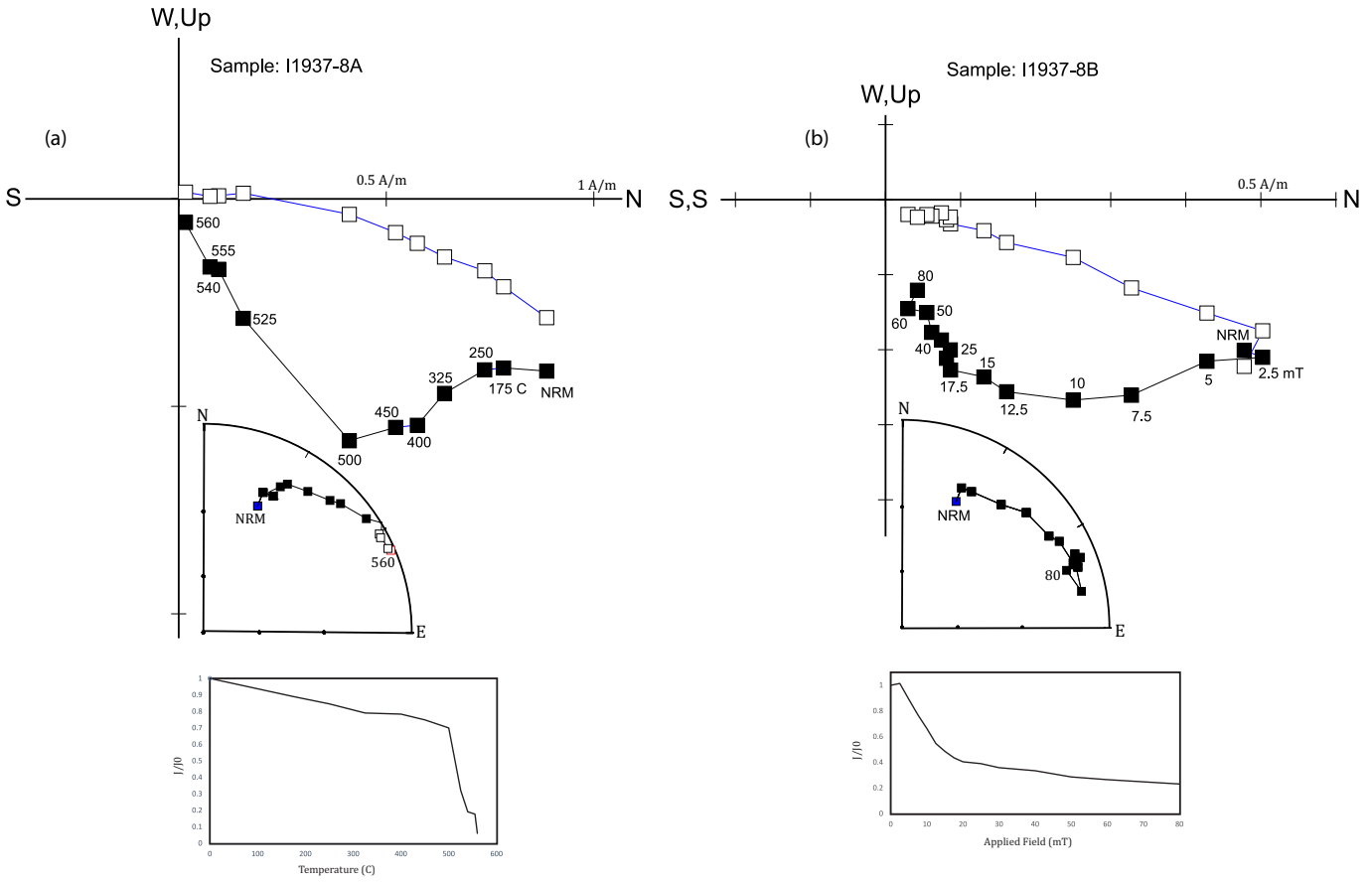


Fig. 3. (a) Pilot thermal demagnetization of sample I1937-8a including a Zijderveld, stereonet and NRM intensity decay plot. The sample shows two components isolated between 0 and 500 °C and 500–560 °C. (b) Pilot alternating field demagnetization of sample I1937-8b including a Zijdeveld, stereo, and intensity decay plot. The two components are less well resolved than in the thermally demagnetized samples.

2010). A total of 30 samples from our four sites yield a mean declination = 70.4 and inclination = +3.8. The virtual geomagnetic pole from these four sites falls at 18.4°N, 175.2°E (K = 114, A95 = 8.6°).

Klootwijk (1974) obtained two slightly different directions from the Gwalior Sills. These include a lower coercivity component directed to the ENE with a shallow positive inclination and a higher coercivity component with a steeper positive inclination (Table 1). Klootwijk (1974) argued that the steeper component was primary. McElhinny et al. (1978) also found slightly higher inclinations when they imposed a blanket 80 mT field on several samples. In our study, samples sometimes exhibit different behavior in AF and thermal demagnetization. In Figs. 3 and 4 we show both AF and thermal partner specimens. Samples from I1937 (Fig. 3) show great circle trends from higher inclinations toward lower inclinations in both thermal and AF. The same is true for samples from I1939 (Fig. 4a, b). Samples from Site I1938 show similar behavior to that described by Klootwijk (1974) and McElhinny et al. (1978) in some AF-treated samples. Thermally treated samples from I1938 move from steeper (northerly) positive inclinations towards more easterly and shallow inclinations and pass through the higher coercivity directions sometimes isolated in the AF-treated samples (Fig. 4e). Samples at site I1938 also contain pyrrhotite which has a high coercivity and an unblocking temperature of ~325 °C (Dekkers, 1990; Dunlop and Özdemir, 2015). During thermal treatment, pyrrhotite commonly converts to magnetite as seen in the temperature-susceptibility diagrams (Fig. 5c; Dunlop and Özdemir, 2015). Overprint directions noted above are slightly steeper than the characteristic direction except for Site I1936 where the overprint is ~4° degrees shallower (Table 1). We conclude that the shallower component represents the characteristic direction in the sills based on the dominant trend towards shallower inclinations at higher temperatures and coercivities.

Previous studies applied a regional tilt-correction to the data using a uniform strike (90°) and dip (4° south; Pradhan et al., 2010). The net effect of this tilt is negligible because the strike is sub-parallel to the observed magnetic declination recorded by the sills. We observed that contacts between the sills and the Morar Formation were irregular but indistinguishable from horizontal, so we do not apply a tilt-correction to our results. Radhakrishna et al. (2013) obtained easterly and shallow directions from three dykes in the Bundelkhand craton that resemble the Gwalior directions (Table 1). Two of their sites (BK25 and BK26) are within ~100 km of the Gwalior sills whilst the third (BK12) is about 160 km from Gwalior. The VGP from BK12 falls outside the VanDamme (1994) cut-off angle and is excluded from our mean result. We include BK25 and BK26 in our compilation which yields a paleomagnetic pole at 13.5 N, 173.7 E (K = 98, A95 = 3.6; B = 17 sites, n = 116 samples; Fig. 7) and places the Bundelkhand craton and the NIB at equatorial latitudes.

Available outcrop near the contact between the sills and the Morar formation was mostly weathered and therefore not amenable to a baked contact test except at Site I222 where a limestone bed is located ~3–4 m above the contact with the sill. The exposed thickness of the sill at I1936 is ~50 m. The NRM in the Morar limestones was weak (average 0.750 mA/m) and demagnetization behavior noisy (Fig. 8a, b), but the mean direction (Dec = 73.8°,

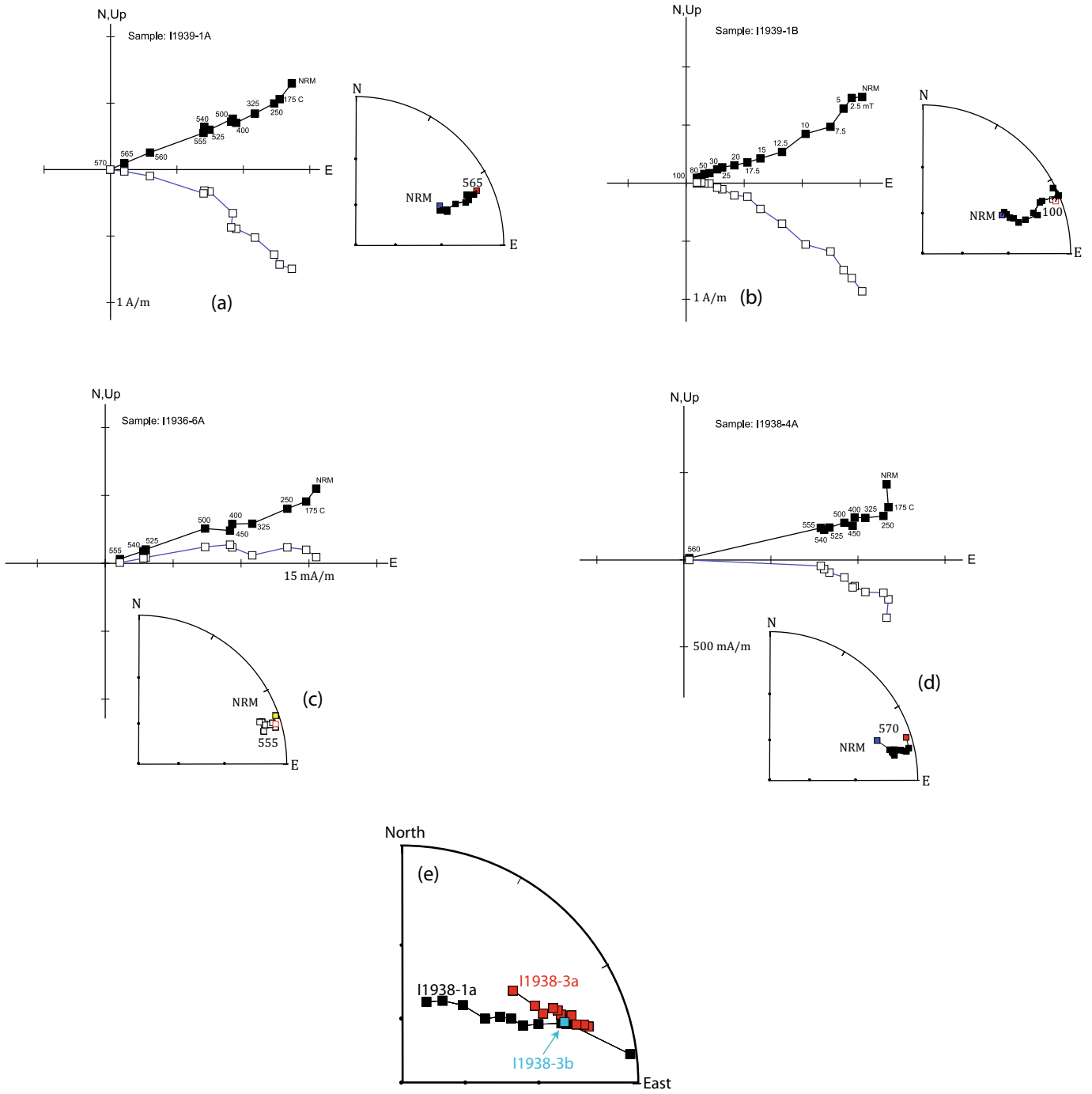


Fig. 4. Representative demagnetization behavior for (a) thermally demagnetized sample I1939-1a; (b) AF-demagnetized sample I1939-1b; (c) thermally demagnetized sample I1936-6a and (d) thermally demagnetized sample I1938-4a. (e) Thermally treated sample great circle trajectories for sample I1938-1a (black) and I1938-3a (red) along with the mean high coercivity direction from sample I1938-3b (blue). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Inc = −19.6°, k = 39, a95 = 10.9°) overlaps (within error) the mean direction at I1936 (Dec = 71.8°, Inc = −10.8°, k = 171, a95 = 4.6°; Fig. 8c). No stable directions were obtained from the Morar Group rocks away from the Gwalior Sills although younger (>1073 Ma) samples from overlying Vindhyan sequence show distinct paleomagnetic directions (Poornachandra-Rao et al., 2005; Malone et al., 2008). We therefore assign a rating of $R4_{C0}$ for the baked contact (Meert et al., 2020).

3. Geochronology

3.1. Geochronology methods

A sample from the mafic sill at site I1939 was processed for U-Pb geochronological analysis at the University of Florida. After crushing, milling, sieving, and decanting, grains <300 μm were separated by density and shape using a Gemini X60 shaking table. This

Table 1
Paleomagnetic Results.

Site	Slat (°N)	Slong (°E)	n/N	Dec°	Inc°	k	a95°	VGPlat (°N)	VGPlong (°E)	Ref
I1936-OP			4/8	76.1	−4.9	123	8.3	—	—	
I1936	26.116	78.079	7/8	71.8	−10.8	171	4.6	13.7	181.3	This study
I1937-OP			6/8	43.8	+12.2	218	4.5	—	—	
I1937	26.232	78.134	7/8	63.0	+1.4	288	3.6	24.4	180.2	This study
I1938-OP			4/9	69.1	+12.1	136	7.9	—	—	
I1938	26.265	78.232	8/9	72.5	+10.4	67	6.8	18.0	171.3	This study
I1939-OP			6/8	73.9	+28.7	91	7.7	—	—	
I1939	26.268	78.228	8/8	74.5	+14.7	131	4.9	17.2	168.3	This study
I438*	26.212	78.167	13/11	81.3	+11.7	89	4.4	10.4	166.7	Pradhan et al., 2010
I439	26.216	78.169	8/11	73.5	+10.8	44	8.5	17.2	170.6	Pradhan et al., 2010
I440	26.210	78.157	12/12	71.8	+0.0	195	3.1	16.3	176.4	Pradhan et al., 2010
I441-A	26.113	78.083	10/14	71.2	−10.3	52	6.7	14.4	181.4	Pradhan et al., 2010
I441-B	26.113	78.083	5/8	72.5	−19.3	588	3.2	11.0	185.0	Pradhan et al., 2010
I441-C	26.113	78.083	6/15	72.9	−3.8	79	7.6	14.4	177.5	Pradhan et al., 2010
*BK-12***	*24.73*	*78.54*	*7/15*	*104.0*	*−14.0*	*16*	*15.5*	*−9.6*	*156.1*	Radhakrishna et al., 2013
BK-25	25.37	78.76	5/13	84	−12.0	25	15.8	2.9	176.8	Radhakrishna et al., 2013
BK-26	25.47	78.14	5/10	83	+7	7	18	7.8	168.0	Radhakrishna et al., 2013
IGWG	26.0	78.0	2/3	85.0	+13.5	119	3.5	7.5	164.0	Klootwijk, 1974
IGWG-HC			5/5	80.0	+39	452	3.9	18.0	151.7	
IGWH	26.0	78.0	1/2	84.6	+16.0	–	–	8.4	163.0	Klootwijk, 1974
IGWH-HC			6/6	77.0	+35	500	3.0	19.6	155.5	
IGWI	26.0	78.0	4/5	69.5	+1.0	106	7.5	18.6	176.8	Klootwijk, 1974
IGWI-HC			6/6	76.0	+35.0	84	7.5	20.5	155.9	
IGWK	26.0	78.0	4/5	79.5	+1.0	318	3.5	9.6	172.2	Klootwijk, 1974
IGWK-HC			6/6	78.5	+29.5	227	4.5	17.0	158.3	
GF-80mT-HC	26.0	78.0	12	78.5	+25.8	48	6.3	16.0	160.5	McElhinny et al., 1978
GT-A	26.0	78.0	7	73.0	+3.0	12	18.0	15.9	174.2	Athavale et al., 1963
I222-Contact	26.116	78.083	6/6	73.8	−19.6	39	10.9	9.8	184.5	This study
Mean-4			**B = 4n = 30**	**70.4**	**+3.8**	**114**	**8.6**	**18.4**	**175.2**	**This study**
Mean-HC	**26.15**	**78.15**	**B = 5, n = 35**	**78.0**	**+33.4**	**K = 495R = 4.99**	**A95 = 3.4**	**18.2**	**156.4**	
Mean Pole	**26.15**	**78.15**	**B = 17n_t = 116**	**75.5**	**+2.2**	**K = 98R = 16.84**	**A95=3.6**	**13.5**	**173.7**	**This compilation**

Notes: Slat = Site latitude, Slong = Site longitude; n = number of samples used in mean; N = number of samples collected; Dec = declination; Inc = inclination; k = Fisher (1953) precision.
parameter; a95 = cone of 95 % confidence about the mean direction; VGPlat = latitude of the virtual geomagnetic pole; VGPlong = longitude of the virtual geomagnetic pole; B = number of sites; n_t = total number of samples; K = Fisher (1953) precision parameter for the paleomagnetic pole; R = length of the resultant vector for the mean paleomagneticpole; A95 = cone of 95 % confidence about the mean of VGP's. HC = high coercivity component. *Pradhan et al., 2010 reported number of specimens used and did not average AF-thermal results from the same samples. **Not used in final mean falls outside the Vandamme (1994) cut-off.

was followed by a magnetic separation using a Frantz LB-1 Magnetic Barrier Laboratory Separator and a final density separation using tetrabromoethane (TBE). The final separate contained an abundance of baddeleyite. The baddeleyite grains were selected for analysis using an optical microscope and mounted in an epoxy resin disk along with baddeleyite from the FC-1 location of the Duluth gabbro. The FC-1 baddeleyite grains used for U-Pb age calibration were provided by George Gehrels (University of Arizona) and are described in detail in Ibañez-Mejia et al. (2014). The baddeleyite grains were imaged using backscatter electron (BSE) and cathodoluminescence (CL) settings on an EVO MA10 XVP scanning electron microscope (SEM) to identify grain structures and zones of metamictization. The grains were ablated using an NWR-ESI 213 nm laser ablation system. To reduce matrix effects, all analyses on FC-1 and unknown baddeleyites were performed with identical laser settings with a 20-μm beam diameter, 10 Hz, 3 % power, and 25-second ablation in a stream of helium (He) at 0.4 ml/minute. Ablated material was mixed with argon (Ar) gas with a flow rate of 1 ml/minute after the ablation cell and the mixture was introduced into the plasma of a Nu-Plasma 3D multi-collector inductively coupled plasma mass spectrometer (LA-MC-ICP-MS). An on-peak gas blank for all masses, including ^{204}Pb + ^{204}Hg, was measured for 20 s before each individual data acquisition ablation. No common Pb correction was applied to the Pb isotopic data. Measurement protocols followed those in Mueller et al. (2008, 2016) with U isotopes measured simultaneously on Faraday collectors and Pb isotopes with ion counters. Two ablations of FC-1 baddeleyite grains bracketed every 10 analyses of the sample baddeleyites. The ablation data were corrected for instrumental drift using the standard-sample-standard spline fit method in the University of Florida CALAMARI spreadsheet using decay constants from Steiger and Jäger (1977) and plotted using IsoplotR (Vermeesch, 2018) and Density Plotter (Vermeesch, 2012). Average and standard deviation values for the 12 FC-1 analyses conducted during data acquisition for this study were 1098.2 Ma ± 1.1 % one sigma and ± 2.3 Ma at the 95 % confidence interval (C.I.).

3.2. Geochronological results

Fifty U-Pb analyses of 26 baddeleyite grains analyzed for this study are provided in a spreadsheet as part of Supplemental File S1. The ablation locations chosen are based on optical, BSE, and CL images, which revealed transparent, light brown, elongate grains (Fig. 9a) with faint magmatic zoning and metamictization (Fig. 9b–c). The images revealed that overgrowths were not apparent and helped to choose ablation sites not compromised by metamict zones or cracks. These data yield ^{207}Pb/^{206}Pb ages from 1675 to 1791 Ma with individual 1σ errors of 2–4 m.y (Table 2; Fig. 9e). These data compare favorably to those in Srivastava et al. (2022), who reported U-Pb ages from seven baddeleyite grains from a Gwalior sill at a different location. Srivastava et al. (2022) used *in situ* SIMS (secondary ion mass spectrometry) that estimated ages ranging from 1777 to 800 Ma. Exclusive of one date of 800 Ma, these data were interpreted to give a ^{207}Pb/^{206}Pb crystallization age of 1733 ± 23 Ma (95 % C.I.) based on the weighted mean of six of the seven analyses reported; however, the weighting factors were not reported. Discordance for these six analyses

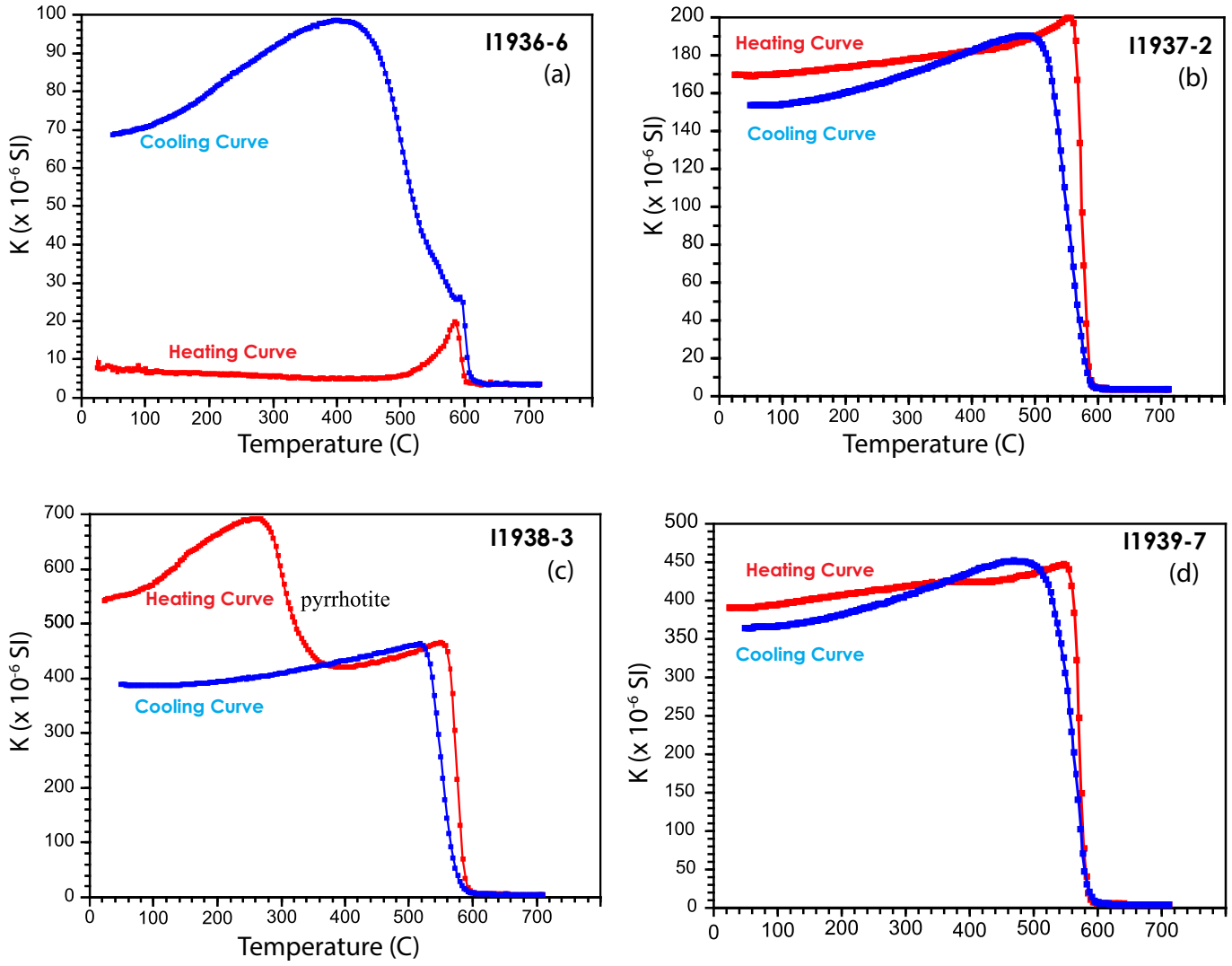


Fig. 5. Temperature-susceptibility response curves for sample (a) I1936-6; (b) I1937-2; (c) I1938-3; (d) I1939-7.

ranged from −18 % to +24 %. The unweighted mean of their data exclusive of the 800 Ma date is 1718 Ma (n = 6).

Data in Table S1 show discordance from −19 % to +7 % in 50 baddeleyite grains from site I1939. There is no evident correlation between percent discordance and age for these analyses (Fig. 9e) and there are no consistent differences in age between the inner and outer parts of individual grains. The positive values for discordance suggest loss of radiogenic Pb, often associated with open system behavior during weathering and/or metamorphism. This is a common feature of datasets from zircon, which also tend to take on nonstoichiometric components during these episodes of open system behavior (e.g., Söderlund et al., 2013; Pidgeon et al., 2023). In contrast, post-crystallization episodes of open system behavior in baddeleyite-bearing rocks are correlated with increases in silica activity and the transformation of baddeleyite to zircon (e.g., Heaman and LeChaminant, 1993; Schaltegger and Davies, 2017; Ibañez-Mejia et al., 2014). We did not detect any grains converted either partially or completely to zircon. The negative values for discordance reflect reverse discordance, a phenomenon not commonly seen in U-Pb studies of zircon. Examples of reverse discordance are more common in U-Pb studies of baddeleyite, including the data reported in Srivastava et al. (2022) for their sample of a Gwalior sill and in several other studies of the U-Pb system in baddeleyite (e.g., Rioux et al., 2010; Ibañez-Mejia et al., 2014; Schaltegger and Davies, 2017).

The lack of correlation of age with discordance, particularly reverse discordance, supports the thesis that the reverse discordance is primarily an indication of intracrystalline diffusion of Pb from high-U/Pb zones to lower U/Pb zones (e.g., Ibañez-Mejia et al., 2014), as opposed to intercrystalline diffusion of common Pb from outside the crystal. Intracrystalline diffusion from high U/Pb zones to lower U/Pb zones inevitably produces apparent relative Pb-loss in some analyses (e.g., Schaltegger and Davies, 2017). When this is the case, the grain has likely acted as a largely closed system, and it is not constructive to arbitrarily remove data with negative or positive discordance values from the age calculation. i.e., true Pb-loss cannot be differentiated from apparent Pb-loss resulting from intracrystalline diffusion. This can be seen in Fig. 9f, which shows a general Gaussian Kernel Density Estimate (KDE) plot (Vermeesch, 2012). This plot uses all fifty ages, their 1 sigma errors and generates a distribution skewed slightly toward younger ages that might reflect some instances of true Pb-loss. The distribution of ages in Fig. 9f shows a single mode at 1717 Ma for n = 50. The entire n = 50 dataset yields an average of 1717 ± 22 Ma (1σ) and ± 6 Ma for the standard error at the 95 % C.I. If a smaller sample set is chosen, e.g., only

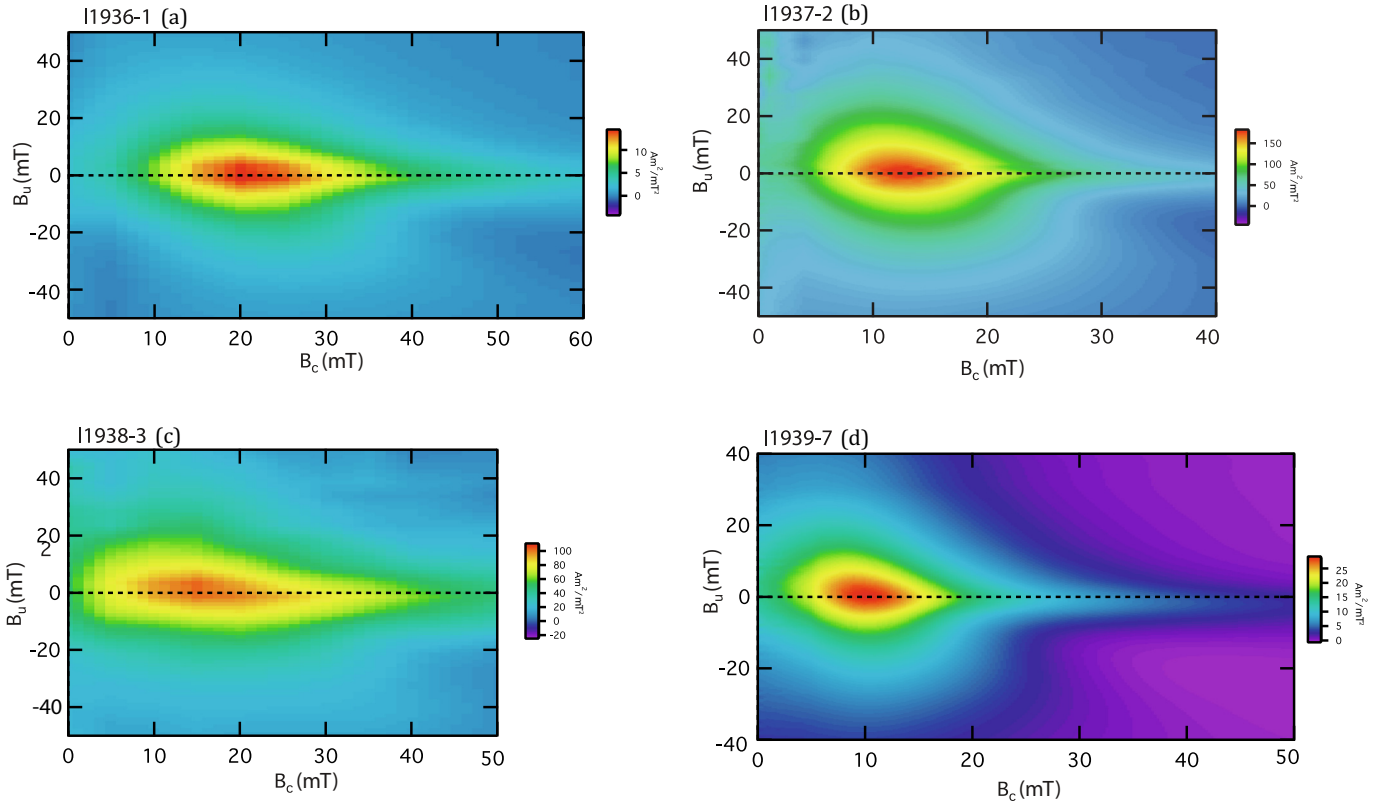


Fig. 6. First-order reversal behavior for sample (a) I1936-1; (b) I1937-2; (c) I1938-3 and (d) I1939-7.

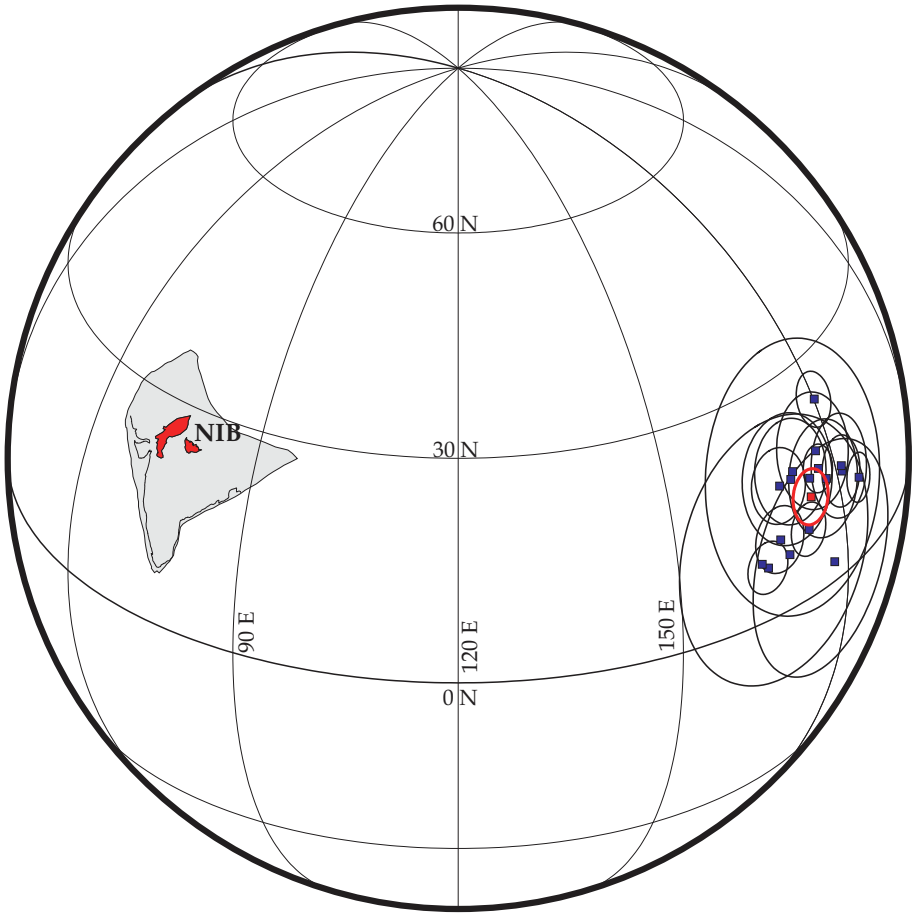


Fig. 7. Virtual geomagnetic poles from this study and previous studies of the Gwalior 'traps' and sills. Mean is denoted in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

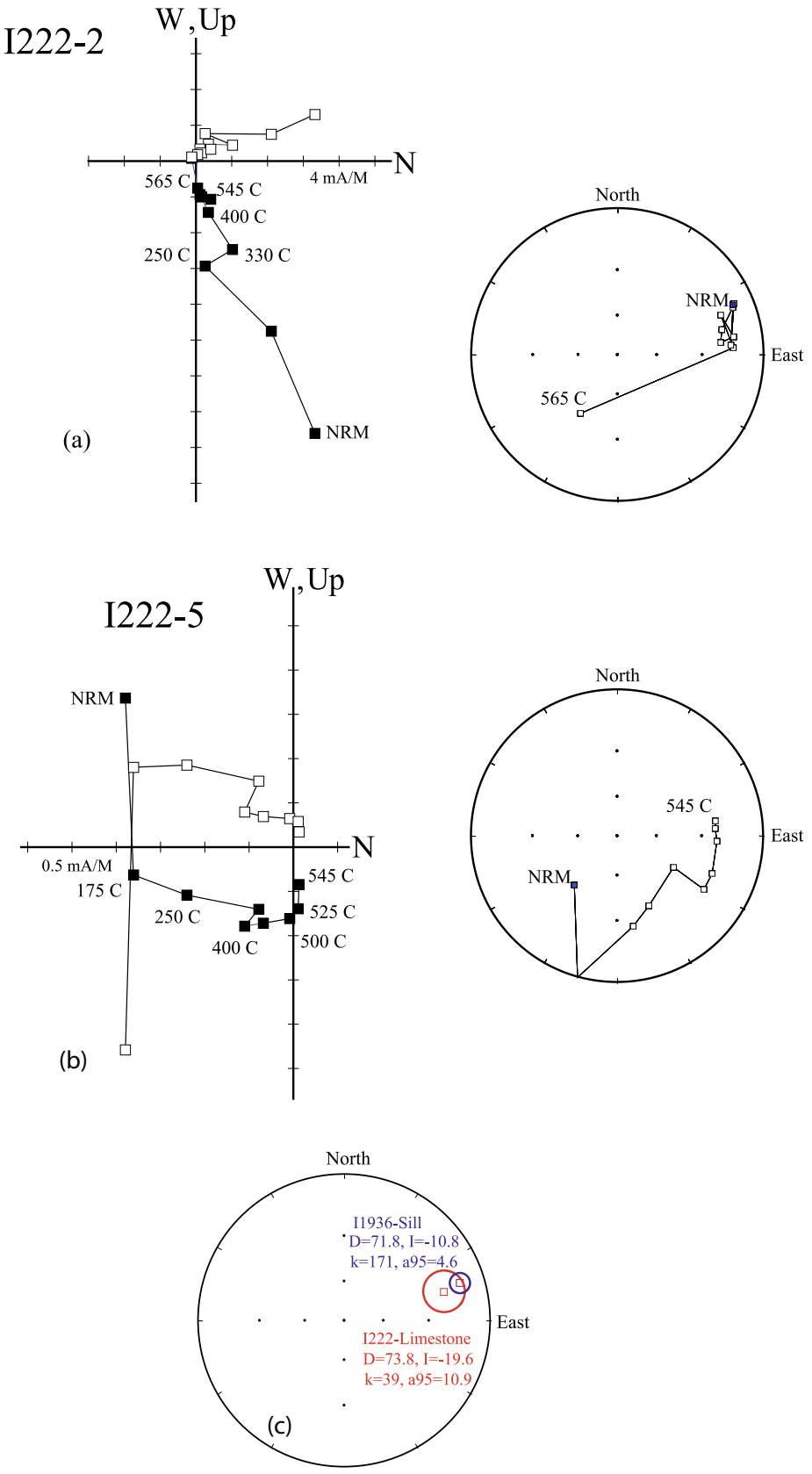


Fig. 8. (a) Orthogonal vector plot of limestone sample I222-2 and stereoplot showing behavior during thermal demagnetization; (b) Orthogonal vector plot of limestone sample I222-5 and stereoplot and (c) comparison of limestone directions to the immediately underlying sill.

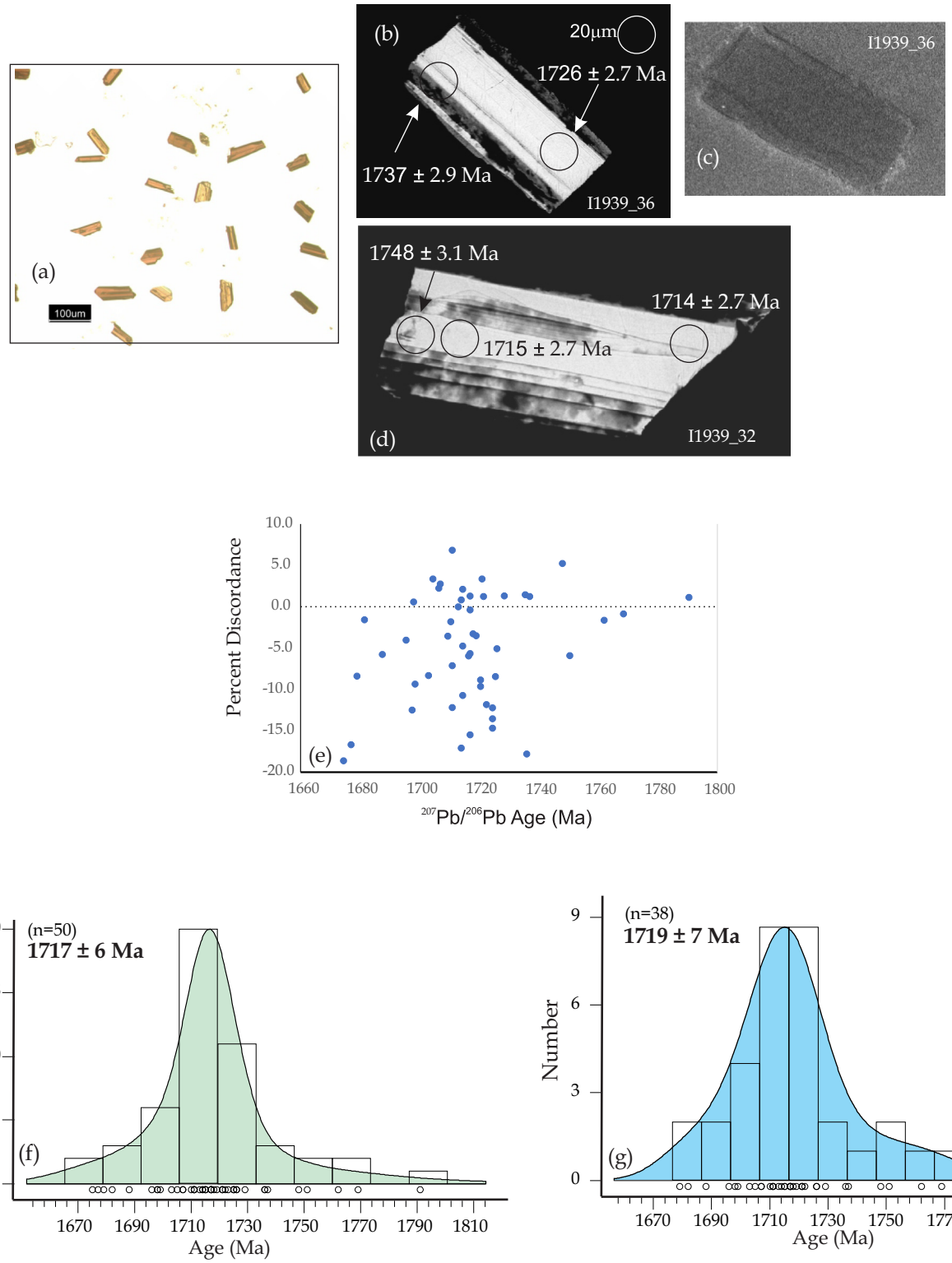


Fig 9. (a) Optical microscopic image of baddeleyite grains (b) Backscattered (BD) image of grain I1939-36 (c) Cathodoluminescent (CL) image of I1939-36 (d) BD image of I1939-32 (e) Percent discordance versus $^{207}Pb/^{206}Pb$ age. (f) A kernel density estimator plot (Vermeesch, 2012) of the $^{207}Pb/^{206}Pb$ ages from all 50 U-Pb analyses of baddeleyites from site I1939 (g) and from the 38 screened analyses used for the age calculation that suggest an age of 1719 ± 9 Ma.

analyses <10 % discordant (n = 38), the average age is 1719 ± 22 Ma (1σ) or 1719 ± 7 Ma (standard error 95 % C.I.; (Fig. 9g). This result is indistinguishable from the age and error derived from the full set at n = 50 (1717 ± 6 Ma). Although culling any of the datasets is arbitrary, the results from all <10 % discordant analyses are likely more robust because the mean is drawn from a more Gaussian distribution, which indicates that parametric statistics are more appropriate. Consequently, our data support that the sill at Site I1939 was emplaced at 1719 ± 7 Ma (95 % confidence).

Table 2
Geochronology Results.

Sample	$^{207}Pb/^{235}U$	1σ (%)	$^{206}Pb/^{238}U$	1σ (%)	$^{207}Pb/^{206}Pb$	1σ (%)	$^{206}Pb/^{238}U$	1σ	$^{*207}Pb/^{235}U$	1σ	$^{207}Pb/^{206}Pb$	1σ	7/6 Discordance	7/5 Discordance
I1939_32_rim1	4.32	7.3	0.2930	7.3	0.10695	0.34	1657	30	1697	30	1748	3.1	5.2	2.4
I1939_32_core	4.37	7.2	0.3018	7.2	0.10499	0.29	1700	54	1706	29	1714	2.7	0.8	0.4
I1939_32_rim2	4.65	7.4	0.3213	7.4	0.10503	0.30	1796	58	1759	30	1715	2.7	−4.8	−2.1
I1939_33_rim	4.48	7.2	0.3102	7.1	0.10480	0.29	1742	54	1728	29	1711	2.7	−1.8	−0.8
I1939_33_core	4.41	7.2	0.3047	7.2	0.10493	0.29	1713	54	1713	29	1713	2.7	0.0	0.0
I1939_34_rim1	4.72	7.2	0.3259	7.2	0.10513	0.30	1819	56	1771	30	1717	2.8	−5.9	−2.7
I1939_34_core	4.79	7.2	0.3339	7.2	0.10410	0.29	1857	58	1784	30	1699	2.6	−9.3	−4.1
I1939_34_rim2	4.75	7.2	0.3189	7.2	0.10815	0.48	1784	56	1777	30	1769	4.3	−0.9	−0.4
I1939_35_core1	4.31	7.7	0.2974	7.7	0.10502	0.33	1678	56	1694	31	1715	3.0	2.1	1.0
I1939_35_core2	4.36	7.4	0.3008	7.4	0.10518	0.31	1695	55	1705	30	1717	2.9	1.3	0.6
I1939_36_core	4.91	7.3	0.3368	7.3	0.10565	0.30	1871	59	1803	30	1726	2.7	−8.5	−3.8
I1939_36_rim	4.47	7.6	0.3049	7.5	0.10631	0.32	1716	57	1725	31	1737	2.9	1.2	0.6
I1939_37_rim	4.19	7.7	0.2912	7.7	0.10445	0.31	1647	56	1673	31	1705	2.9	3.3	1.5
I1939_37_core	4.24	7.4	0.2938	7.4	0.10460	0.32	1660	54	1681	30	1707	2.9	2.7	1.2
I1939_38_rim1	4.56	7.2	0.3161	7.2	0.10474	0.32	1770	56	1743	30	1710	2.9	−3.6	−1.6
I1939_38_core	4.42	7.6	0.3029	7.6	0.10582	0.32	1706	57	1716	31	1729	2.9	1.3	0.6
I1939_38_rim2	5.17	7.1	0.3549	7.1	0.10559	0.34	1958	60	1847	30	1725	3.1	−13.5	−6.0
I1939_39_core1	4.06	7.3	0.2806	7.3	0.10484	0.36	1594	51	1645	29	1711	3.3	6.8	3.1
I1939_39_core2	4.94	7.1	0.3399	7.1	0.10537	0.32	1886	58	1809	30	1721	2.9	−9.6	−4.3
I1939_40_core	4.77	7.4	0.3313	7.4	0.10437	0.29	1845	59	1779	31	1703	2.6	−8.3	−3.7
I1939_41_core	4.95	7.1	0.3448	7.1	0.10406	0.29	1910	59	1810	30	1698	2.7	−12.5	−5.5
I1939_41_rim	4.74	7.5	0.3249	7.5	0.10569	0.34	1814	59	1773	31	1726	3.1	−5.1	−2.3
I1939_42_core (l)	4.63	7.2	0.3262	7.2	0.10301	0.28	1820	57	1755	30	1679	2.6	−8.4	−3.7
I1939_42_core (u)	4.55	7.1	0.3190	7.1	0.10350	0.30	1785	55	1741	29	1688	2.7	−5.8	−2.5
I1939_43_rim	4.61	7.1	0.3179	7.1	0.10529	0.31	1779	55	1752	29	1719	2.9	−3.5	−1.6
I1939_43_core	5.29	7.5	0.3653	7.4	0.10499	0.34	2007	64	1867	31	1714	3.1	−17.1	−7.5
I1939_44_core1	5.11	7.1	0.3609	7.1	0.10276	0.28	1986	60	1838	30	1675	2.6	−18.6	−8.1
Sample	**$^{207}Pb/^{235}U$**	**1σ (%)**	**$^{206}Pb/^{238}U$**	**1σ (%)**	**$^{207}Pb/^{206}Pb$**	**1σ (%)**	**$^{206}Pb/^{238}U$**	**1σ**	**$^{*207}Pb/^{235}U$**	**1σ**	**$^{207}Pb/^{206}Pb$**	**1σ**	**7/6 Discordance**	**7/5 Discordance**
I1939_44_core2	5.03	7.0	0.3547	7.0	0.10292	0.29	1957	59	1825	29	1677	2.6	−16.7	−7.2
I1939_45_core1	4.75	7.6	0.3288	7.6	0.10483	0.36	1833	60	1777	31	1711	3.3	−7.1	−3.2
I1939_45_core2	5.01	7.3	0.3468	7.3	0.10482	0.34	1919	60	1821	31	1711	3.2	−12.2	−5.4
I1939_52_core	4.30	7.3	0.2994	7.2	0.10409	0.31	1688	54	1693	29	1698	2.9	0.6	0.3
I1939_52_rim	4.45	7.1	0.3066	7.1	0.10518	0.32	1724	53	1721	29	1717	2.9	−0.4	−0.2
I1939_53_core	4.28	8.0	0.2944	8.0	0.10539	0.30	1663	58	1689	32	1721	2.7	3.3	1.5
I1939_53_rim	4.60	7.6	0.3168	7.6	0.10521	0.31	1774	59	1749	31	1718	2.8	−3.3	−1.5
I1939_54_core1	5.23	7.3	0.3590	7.2	0.10561	0.32	1978	61	1857	30	1725	2.9	−14.7	−6.5
I1939_54_core2	4.92	7.1	0.3333	7.1	0.10710	0.42	1854	57	1806	30	1751	3.9	−5.9	−2.7
I1939_55_core	4.32	7.2	0.3034	7.2	0.10316	0.29	1708	53	1696	29	1682	2.7	−1.6	−0.7
I1939_56_core	5.10	7.0	0.3502	7.0	0.10558	0.34	1935	59	1836	29	1725	3.1	−12.2	−5.4
I1939_57_core1	5.22	7.1	0.3602	7.1	0.10517	0.31	1983	60	1856	30	1717	2.8	−15.5	−6.8
I1939_57_core2	4.51	7.1	0.3148	7.1	0.10395	0.30	1764	55	1733	29	1696	2.7	−4.0	−1.8
I1939_58_rim	4.90	7.3	0.3371	7.3	0.10537	0.30	1873	59	1802	30	1721	2.7	−8.9	−3.9
I1939_58_core	5.07	7.4	0.3483	7.4	0.10550	0.33	1926	62	1830	31	1723	3.0	−11.8	−5.2
I1939_29_core	4.39	7.1	0.3018	7.1	0.10543	0.31	1700	53	1710	29	1722	2.9	1.2	0.6
I1939_60_core1	4.45	7.1	0.3039	7.1	0.10623	0.32	1710	53	1722	29	1736	2.9	1.4	0.7
I1939_60_core2	5.47	7.2	0.3733	7.2	0.10625	0.31	2045	63	1896	31	1736	2.8	−17.8	−7.9
I1939_62_core	4.71	7.1	0.3251	7.1	0.10518	0.31	1814	56	1770	29	1717	2.9	−5.7	−2.5
I1939_24_core1	4.76	7.4	0.3202	7.4	0.10779	0.39	1791	58	1778	31	1762	3.5	−1.6	−0.7
I1939_24_core2	4.77	7.3	0.3161	7.3	0.10949	0.68	1770	56	1780	30	1791	6.2	1.1	0.5
I1939_25_core	4.26	7.1	0.2954	7.1	0.10457	0.28	1669	52	1685	29	1707	2.6	2.2	1.0
I1939_26_core	4.96	7.1	0.3424	7.1	0.10503	0.27	1898	58	1812	29	1715	2.5	−10.7	−4.7

Analytical errors are reported at the 1-sigma (σ) level in percent for isotopic ratios and 1-sigma in Ma for ages. 7/6 discordance calculated as [1-($^{206}Pb/^{238}U$age/$^{207}Pb/^{206}Pb$ age)*100] and.
7/5 discordance is calculated as [1-($^{206}Pb/^{238}U$age/$^{207}Pb/^{235}U$ age)*100].

4. Discussion

4.1. South India-Northern India consolidation

Meert et al. (2021) provide an up-to-date listing of the Proterozoic key poles from both the SIB and NIB. Based on our geochronological data and those of Srivastava et al. (2022), we use an age of ~1.725 Ga for our reconstructions and analysis of the Gwalior Sills. There are no strictly coeval poles from the NIB and SIB prior to 1.1–1.0 Ga. Pivarunas et al. (2021) presented new data and a new grand mean pole for 1.765 Ga dykes from the Singhbhum craton (SIB) which are ~35 Ma older than the Gwalior results. The paleomagnetic pole from the Singhbhum craton falls at 43.3° S, 139.8°E (A95 = 10.3°) and is statistically distinct from the Gwalior pole reported herein at 13.5°N, 173.7°E. Depending on the choice of polarity, the two poles are separated by either 65 or 115°. Mean rates of apparent polar wander (APW) range from 1.45° to 2.5° Ma^{-1} which is 3–5 times the background APW rate for the past 320 Ma (Torsvik et al., 2012). Despite the significant difference in paleomagnetic pole positions, both the Singhbhum (+Dharwar and Bastar) craton and the Bundelkhand craton were at low paleolatitudes between 1.765 and 1.719 Ga (Fig. 10a). Miller et al.

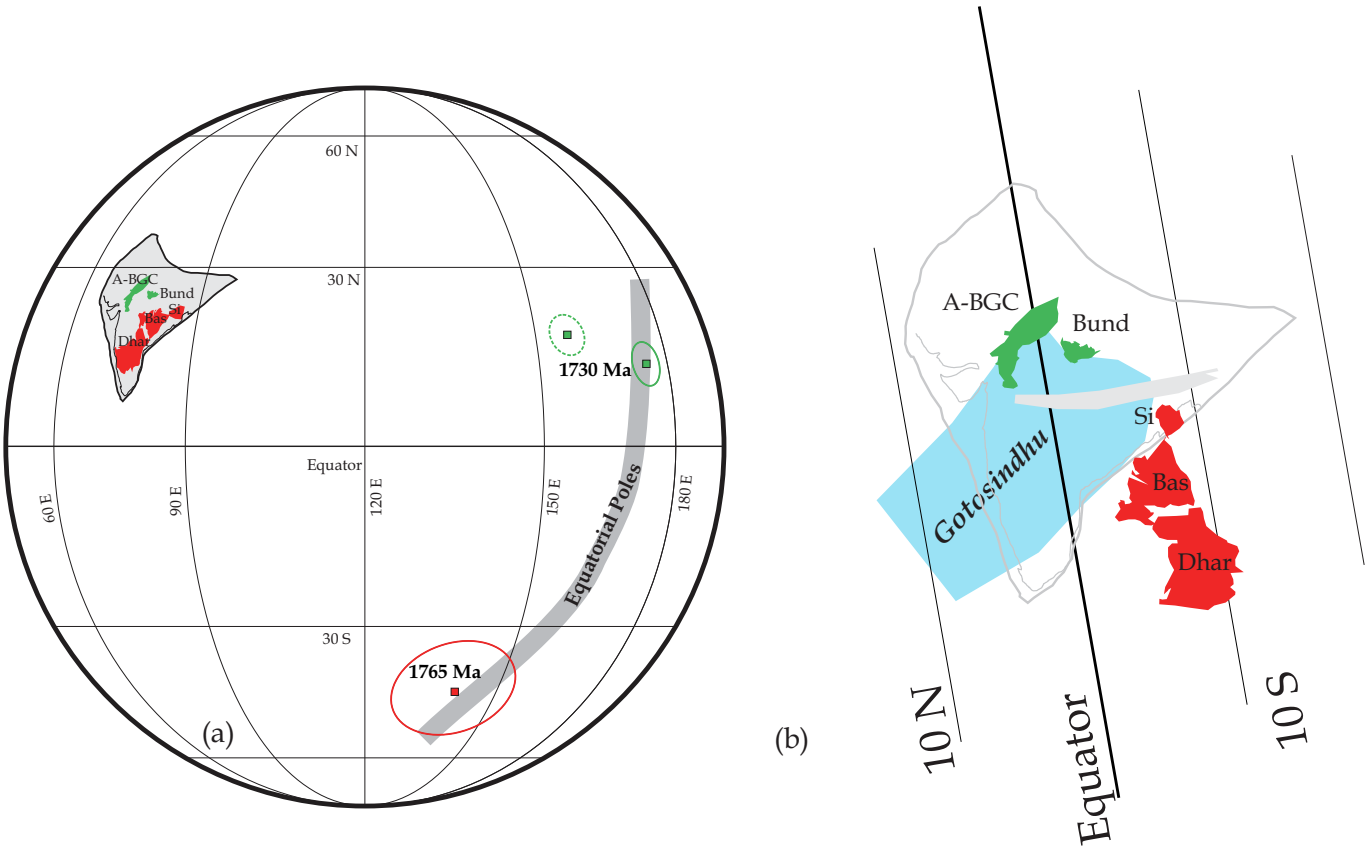


Fig. 10. (a) Paleomagnetic poles from the Gwalior sills (this study) and 1765 Ma Newer dolerites (Pivarunas et al., 2021; Shankar et al., 2017). Grey band represents a locus of theoretical paleomagnetic poles that would place Gwalior, India (26 N, 78 E) at equatorial latitudes. (b) Possible reconstruction of the North Indian Blocks (NIB) and South Indian Blocks (SIB) at ~1750 Ma and the position of the proposed Gotosindhu. Red circle represents the Newer dolerites pole and A95. Solid green circle is the Gwalior sills pole. Dashed green circle is the Gwalior sills VGP as calculated from the steep directions (see Table 1). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(2023) also provide two VGPs from slightly older, ~1.8 Ga dykes in the Dharwar craton that also indicate low paleolatitudes.

Fig. 10a shows the paleomagnetic poles from Gwalior and Singhbhum cratons (plotted as south poles) and the present-day geographic relationships between the NIB, the SIB and the CITZ. Fig. 10b shows the position of the NIB (Aravalli-BGC and Bundelkhand) using the Gwalior pole and the SIB (Dharwar, Bastar and Singhbhum) using the Singhbhum pole. Since longitude is unconstrained, we have placed the blocks in their 'closest approach configuration' (Meert and Torsvik, 2003). Fig. 10b is schematic, and the SIB/NIB could also be quite distant from one another. One possibility is that the basin originated during rifting (Singh and Chakraborty, 2021) between the NIB and an unknown conjugate around 1750 Ma which opened a sizeable ocean basin. We have chosen the name Gotosindhu after the Bengali words "*Goto*" meaning 'lost or departed' and '*Sindhu*' (meaning ocean or sea). Closure of the Gotosindhu commenced around 1.65 Ga (Bhowmik et al., 2012a, 2012b; Bhowmik and Chakraborty, 2017; Bhowmik, 2019) via subduction beneath the SIB with a locus near the CITZ. Back-arc spreading due to slab roll-back took place between 1.62 and 1.60 Ga and an arc collision with the SIB took place shortly thereafter (Bhowmik and Chakraborty, 2017). Subduction ceased around 1.52 Ga after the closure of the back-arc systems with the SIB. These arc collisions did not result in the unification of the NIB and SIB, but likely caused the closure of sedimentation in the Lower Vindhyan basin (Bhowmik, 2019; Chattopadhyay et al., 2020; Banerjee et al., 2021).

We have argued (Meert et al., 2021; Miller et al., 2023) along with others (Bhowmik, 2019; Chattopadhyay et al., 2020; Banerjee et al., 2021; Ghosh et al., 2022) that the widespread 1.1–0.93 Ga deformation throughout central India resulted in the final amalgamation of Peninsular India through the collision of the SIB and NIB along the CITZ. This orogenesis coincides with the formation of the Rodinia supercontinent. At present, there are no paleomagnetic data to fully evaluate this model; however, the new data reported in this study along with 1.06–0.93 Ga paleomagnetic data reported in Meert et al. (2021) are consistent with the starting and ending points of the Bhowmik (2019) model.

4.2. India in the Columbia supercontinent

Meert et al. (2021) and Meert and Santosh (2017, 2022) provided detailed reviews of India's reconstructed position in the Columbia supercontinent. Nearly every model places Peninsular India along the periphery of the Columbia supercontinent. There is no agreement on the exact position or orientation of India. Zhao et al. (2002) placed India in between Australia and North China and aligned the CITZ with the Trans-North China Orogen (TNCO). Zhang et al. (2012) place India adjacent to North China, Siberia, and West Africa (Fig. 11a). Pisarevsky et al. (2014) sandwich India between Congo-Sao Francisco, Kalahari and Baltica (Fig. 11b,c). Chaves and deRezende (2019) position India between Siberia and proto-Australia. Each of those reconstructions shows Peninsular India as a single block. Paleomagnetic data are scarce, and geological support for these published positions is also problematic.

Meert et al. (2021) note that the CITZ-TNCO connection used to link North China and India is based on assigning a 2.1–1.9 Ga age

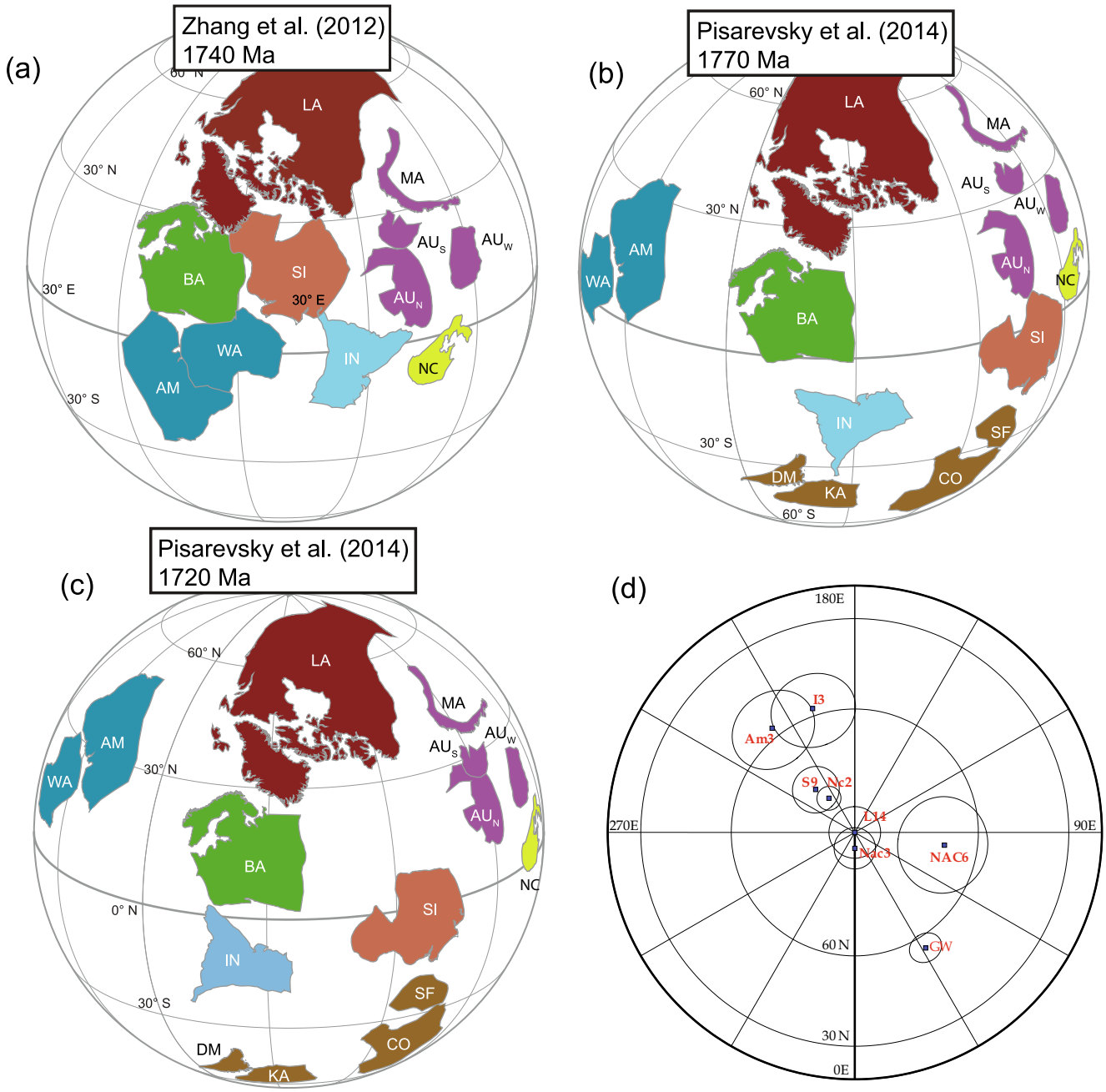


Fig. 11. Models of the early phases of Columbia supercontinent assembly according to (a) Zhang et al. (2012) at 1740 Ma; (b) Pisarevsky et al. (2014) at 1770 Ma; (c) Pisarevsky et al. (2014) at 1720 Ma. Craton/Continent abbreviations: AM: Amazonia, AU_N: North Australia, AU_S: South Australia, AU_W: West Australia, BA: Baltica, CO: Congo, DM: Dronning Maud Land, KA: Kalahari, IN: India, NC: North China, SF: Sao Francisco, SI: Siberia, WA: West Africa, LA:Laurentia, MA:Mawson. (d) Paleomagnetic poles used by Zhang et al. (2012) along with the Gwalior and Newer Dolerites pole. Poles were rotated using the Euler pole data listed in Zhang et al. (2012).

range for high-grade metamorphism and deformation in the CITZ. There is little evidence for widespread orogenesis within the CITZ at that time although localized UHP metamorphism is documented in the Bhandara-Balaghat Granulite domain at 2.05 Ga ((Bhowmik et al., 2005). Pisarevsky et al. (2014) argue for a Baltica-India connection and correlate the CITZ with the Pachelma orogenic belt separating Volgo-Uralia and Sarmatia which is also assigned to the interval 2.1–1.9 Ga. In their model, India is also sandwiched between the Congo-Sao Francisco and Kalahari cratons. Because these models align the orogenic belts along the strike, they need not reflect a cycle of amalgamation between the CITZ and either Baltica or North China. Zhang et al. (2012) position India between N. China, NW Australia, Siberia, and West Africa like the Chaves and deRezende (2019) reconstruction but provide no strong geological argument for that location.

It is not surprising that India's position within Columbia is contentious. With the addition of the Gwalior pole, there are now only three dated poles that fall broadly within the lifespan of Columbia assembly and breakup (~1.75–1.40 Ga). We review the Zhang et al. (2012) model at 1.75 Ga using the Gwalior and Newer dolerites poles following the Bayesian statistical methods outlined in Heslop and Roberts (2019) and Meert and Santosh (2022). Each pole is rotated into the proposed fit and then evaluated for goodness of fit using the Bayes error. A Bayes error = 0.5 indicates that both poles are identical whereas a Bayes error = 0 indicates both poles are different. Heslop and Roberts (2019) did not consider

age differences between poles which limit the utility of their approach in the Proterozoic; however, Meert and Santosh (2022) modified the analysis to account for age differences between poles. Meert and Santosh (2022) argued that pole-to-pole comparisons with Bayes errors < 0.2 should be viewed with caution in making reconstructions. Misfits between poles indicates errors in orientation (rotation), latitude, or both.

The Zhang et al. (2012) model (Fig. 11a) places India next to North China, Siberia, and West Africa. Fig. 11d shows all poles used in that reconstruction with our new Gwalior pole and the Newer dolerite 1.765 Ga pole (Pivarunas et al., 2021; Shankar et al., 2017) which were not available at the time of the Zhang et al. (2012) publication. There were no data at the time for West Africa and Siberia, but there were data from Amazonia (to which West

Table 3
Paleomagnetic Poles Used in Fig. 12.

Pole Name	Craton	Nominal Age (Ma)	Plat	Plon	A95°	Reference
1. Para de Minas	Congo-SF	1796	39.8°S	196.8°E	17°	(D'Agrella-Filho et al., 2020)
2. AM4-Avanavero	Amazonia	1789	48.4°N	207.9°E	9.2°	Reis et al. (2013); Bispos-Santos et al. (2014)
3. ECMB	Laurentia	1779	20.4°N	265.8°E	4.5°	Swanson-Hysell et al. (2021)
4. B9-Shoksa Fm	Fennoscandia	1775	39.7°N	221.1°E	4.0°	Pisarevsky and Sokolov (2001)
5. NC2-Taihang	North China	1769	36.0°S	067.0°E	2.8°	Halls et al. (2000); Xu et al. (2014)
6. NC3-Yinshan	North China	1769	32.3°S	068.3°E	2.0°	Halls et al. (2000); Xu et al. (2014)
7. I3-Newer Dolerites	South India	1765	43.0°N	320.0°E	10.3°	(Pivarunas et al., 2021); Shankar et al. (2017)
8. NAC2- Elgee	N. Australia	1762	5.4°S	211.8°E	3.2°	Schmidt and Williams (2008)
9. L13-Jan Lake Granite	Laurentia	1758	28.1°N	268.9°E	17.2°	Gala et al. (1995)
10. B11- Korosten C/D	Sarmatia	1755	26.5°N	169.1°E	3.9°	Elming et al. (2010)
11. B10-Ropruchey	Fennoscandia	1751	39.1°N	216.6°E	6.7°	Fedotova et al. (1999), Lubnina et al. (2012)
12. Igueda-VGP	W. Africa	1747	4.0°S	262.1°E	2.5°	Neres et al. (2016)
13. L14-Cleaver	Laurentia	1740	21.1°N	276.7°E	6.2°	Irving et al. (2004)
14. S4-Elgetei	Siberia	1732	7.1°N	183.5°E	13.2°	Didenko et al. (2015)
15. Gwalior	North India	1730	13.5°S	353.7°E	3.6°	This study
16. NAC3-Peters Creek	N. Australia	1727	26.0°S	221.0°E	4.8°	Idnurm (2000)

Notes: Plat = Pole latitude; Plong = Pole Longitude; A95 = cone of 95 % confidence about the mean pole calculated from VGP's. Labels (e.g. B10 etc) refer to the compilation of poles in Elming et al. (2021). Rotation Parameters (all poles rotated to North): Laurentia 0°N, 180.5°E, +66.7°; Fennoscandia 32.3°N, 144.4°E, +61.1°; Volgo-Uralia 32.3°N, 144.4°E, +61.1°; Sarmatia 32.3°N, 144.4°E, +61.1°; Amazonia 65.2°N, 172.9°E, +115.1°; N. Australia 0°N, 303.1°E, −98.5°; South Indian Blocks 16.7°N, 237.3°E, +49.2°; North China 20.6°N, 112.6°E, −141.2°; Siberia 34°N, 130°E, +105.9°; North India 0.7°N, 82.0°E, −104.4°; W. Africa 31.8°N, 192.2°E, +108.5°; Congo 22.1°, 226.8°E, −155.5°; Sao Francisco 34.8°N, 195.9°E, −150.7°.

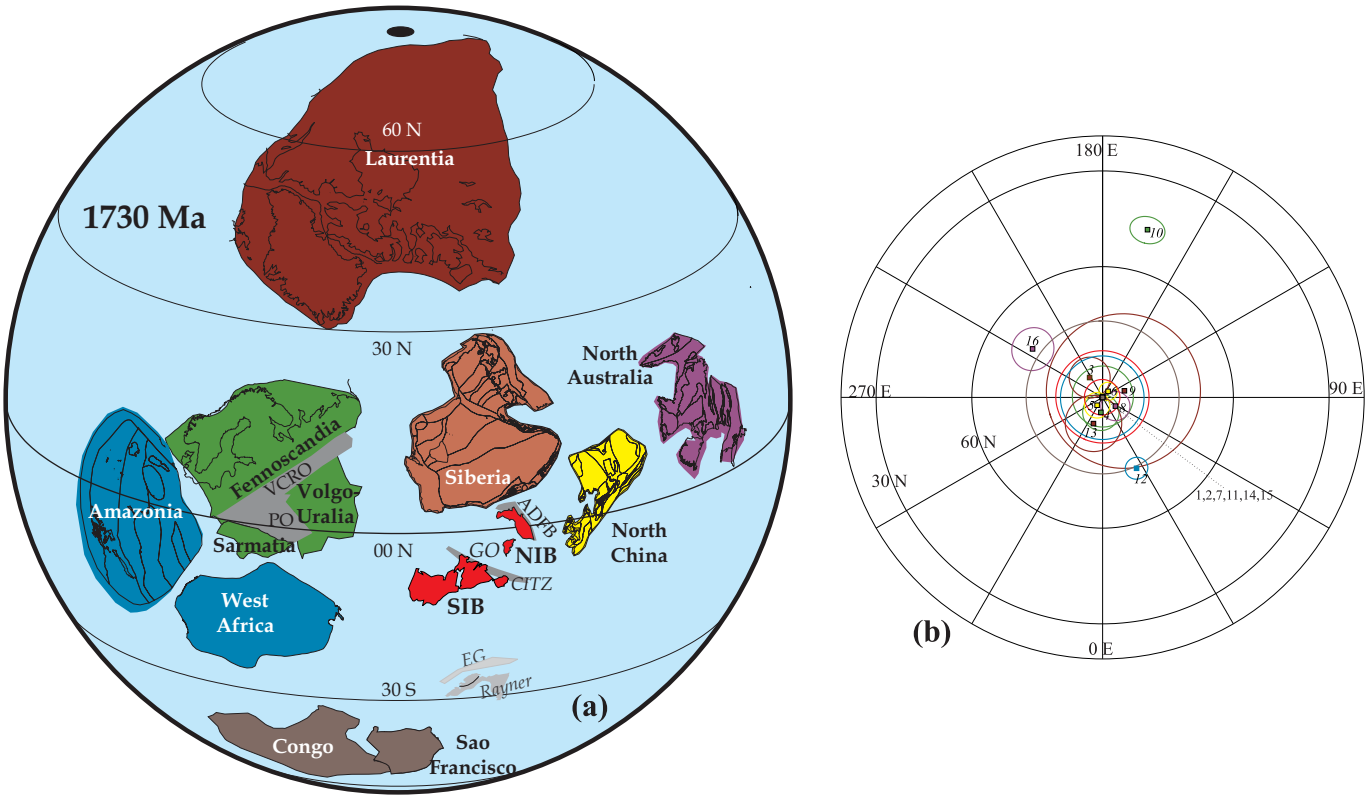


Fig. 12. (a) Possible reconstruction at ~1730 Ma using the poles and rotational parameters listed in Table 3 and (b) Paleomagnetic poles in Table 3 rotated to the North pole. GO = Gotosindhu; 1.65–1.55 Ga orogenic belts: EG = Eastern Ghats Mobile Belt; CITZ = Central Indian Tectonic Zone; ADFB = Aravalli-Delhi Fold Belt; Orogenic Belts in Baltica PO = Pachelma Orogen (2.0–1.9 Ga); VCRO = Volhyn-Central Russia Orogen (1.9–1.75 Ga).

Africa is affixed) and North China. Neither the Gwalior nor the Newer Dolerite pole matches well with the North China craton pole with Bayes errors ranging from 0.0 to 0.007. Only the Newer dolerites pole exceeds the 0.2 Bayes error threshold (after age adjustment) with Amazonia.

Pisarevsky et al. (2014) position India near Sarmatia-Volgo Uralia from 1.77 to 1.27 Ga. This model was based on a compatible position between Baltica-India at ~1.465 Ga which, at the time, was the only pole available from India for use in Columbia models. The Newer dolerites pole (Shankar et al., 2017; Pivarunas et al., 2021) is compatible with this reconstruction although there are age differences between the CITZ and Pachelma orogens as noted above (Fig. 11b). (Elming et al., 2021) use the same position for India in their 1.78 and 1.71 Ga reconstructions.

We do not reject any of these proposals for the position of the Indian blocks. Zhang et al. (2012) were prescient in assuming an equatorial position for the Indian blocks in the 1.765–1.725 Ga interval. The misfit noted above is the result of rotational discrepancies in the reconstruction. Pisarevsky et al. (2014) created their model based on a single 1.465 Ga pole from the SIB and were also remarkably prophetic about the equatorial position of the Indian blocks during the 1.765–1.74 Ga interval. Table 3 contains a listing of paleomagnetic poles from around the globe with ages spanning ~1.79–1.725 Ga. Fig. 12a shows the possible relationships between various cratonic regions based on the data in Table 3 with the acknowledgment of limitations on the paleomagnetic data including hemispheric ambiguity and longitudinal freedom. Fig. 12b shows the paleomagnetic poles rotated into this configuration. Poles 10 (Sarmatia) and 15 (Australia) are outliers whereas the other poles indicate a reasonable fit in the proposed reconstruction. Pole 12 (West Africa) is also displaced but represents a VGP in which secular variation is not averaged. We position the NIB and SIB between Baltica, Siberia, and North China, but longitude is not constrained in this reconstruction. Dasgupta et al. (2017) explored the complex history of the Eastern Ghats Mobile Belt (EGMB). The polyphase metamorphism and tectonism within the various domains of the EGMB hinder a coherent model of collisional orogenesis (Sawant et al., 2017; Meert et al., 2017). Recent reviews and geophysical investigations suggest that much of the EGMB developed separately from the SIB and is better viewed as part of the Rayner Province (East Antarctica) that was later attached to the SIB during the 1.0–0.9 Ga period or during Gondwana assembly (Bose et al., 2021; Giri et al., 2022). There are no reliable paleomagnetic data from the Rayner Province during the Mesoproterozoic, so we show the EGMB attached to the Rayner Province in our reconstruction separate from both the NIB and SIB (Fig. 12a).

5. Conclusions

We report an updated paleomagnetic pole from the Gwalior sills in the Bundelkhand craton of northern India which includes four additional sampling sites. The pole falls at 13.5°N, 173.7°E (A95 = 3.6°, K = 98, n = 17 sites).

We report a new $^{207}Pb/^{206}Pb$ baddeleyite age of 1719 ± 7 Ma (95 % C.I.) for the Gwalior sill at Site I1939 which is slightly younger than the previous age determinations.

Limestone samples taken 3–5 m above the upper contact with the sill at Site I1936 yielded stable directions that were statistically indistinguishable from the sill directions at that location, based on overlapping a_{95} envelopes. Samples more distant from the contact were unstable so we rate the baked contact test as $R4_{C0}$. The pole does not match any younger poles from the NIB, and we tentatively conclude that it records a magnetization dating to the emplacement of the sills. The new Gwalior pole receives an R = 5 rating based on the Meert et al. (2020) criteria as it lacks reversals and does not average secular variation according to the Deenan et al. (2011, 2014) parameters ($A95_{min}$ = 3.9° > $A95_{GWAL}$ = 3.6°).

The South Indian Block (SIB) and North Indian Block (NIB) were both at low paleolatitudes during 1719–1765 Ma, but not in their present-day configurations. We postulate that the Gotosindhu, an ancient sea, separated them at 1730 Ma and that the final amalgamation along the CITZ did not take place until 1.1–1.0 Ga although models positing a close approach of the NIB/SIB around 1.6 Ga remain plausible.

The position of the Indian blocks within Columbia remains enigmatic. Previous models can accommodate the data reported herein but require rotational adjustments and reconsideration of geological connections.

Orogenesis in the CITZ is related to the assembly of Peninsular India and should not be considered part of the Columbia-forming orogenic belts (Bhowmik, 2019; Banerjee et al., 2021). The Aravalli-Delhi orogeny may be correlated with Columbia assembly; however, the current age constraints span from 1.8 to 0.95 Ga and lack detailed geological correlations elsewhere in Columbia (Biswal et al., 2022; Banerjee et al., 2021).

CRediT authorship contribution statement

Joseph G. Meert: Conceptualization, Methodology, Data curation, Formal analysis, Funding acquisition, Investigation, Writing - original draft, Writing - review & editing. **Scott.R. Miller:** Conceptualization, Investigation, Writing - review & editing. **Anthony Pivarunas:** Investigation, Writing-review & editing. **Manoj K. Pandit:** Conceptualization, Writing original draft, Writing-review and editing. **Paul Mueller:** Conceptualization, Writing-review and editing, Formal Analysis. **Anup K Sinha:** Writing-review and editing. **George Kamenov:** Formal Analysis, Writing-review & editing. **Samuel Kwafo:** Writing-review & editing. **Ananya Singha:** Writing-review & editing, formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.gr.2023.08.004.

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